Transformer Basics: "Attention is All You Need"

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https://dannagurari.colorado.edu/course/neural-networks-and-deep-learning-spring-2026/

Review

- Last week:
  - Mystery: why did neural networks struggle to translate long sentences?
  - Idea: instead of summarizing everything, attend to relevant inputs
  - Beyond text: attention for image captioning
  - Beyond vision-language problems: attention for speech recognition
- Assignments (Canvas):
  - Problem set 2 due Friday
  - Problem set 3 due in a week
- Questions?

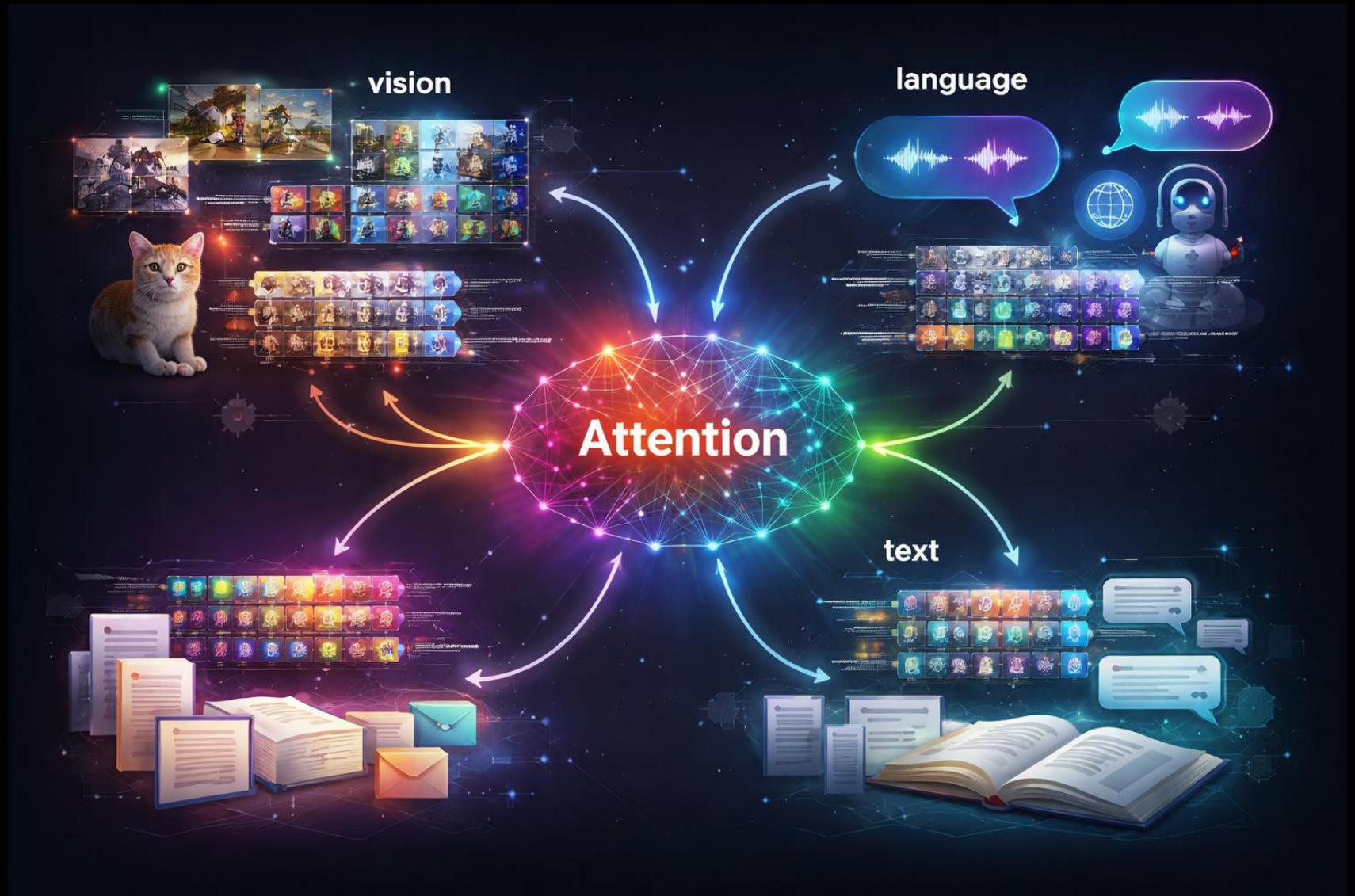
Today’s Topics

- Attention transformed: birth of a new architecture
- Key ingredient: self-attention
- Building a transformer
- Pioneering model for machine translation

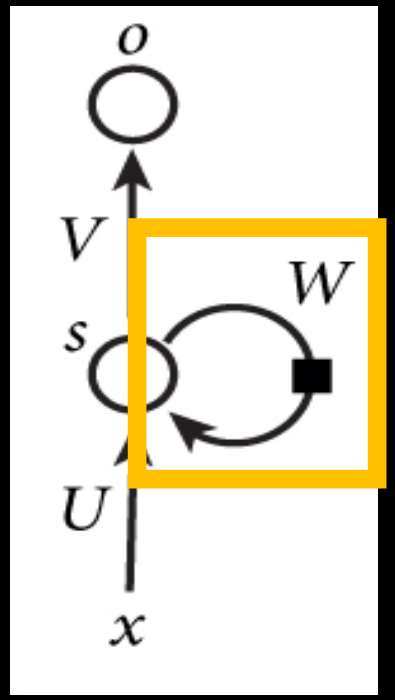
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- Attention transformed: birth of a new architecture
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Context: Across Domains, Attention Emerged as a Key Component for Success



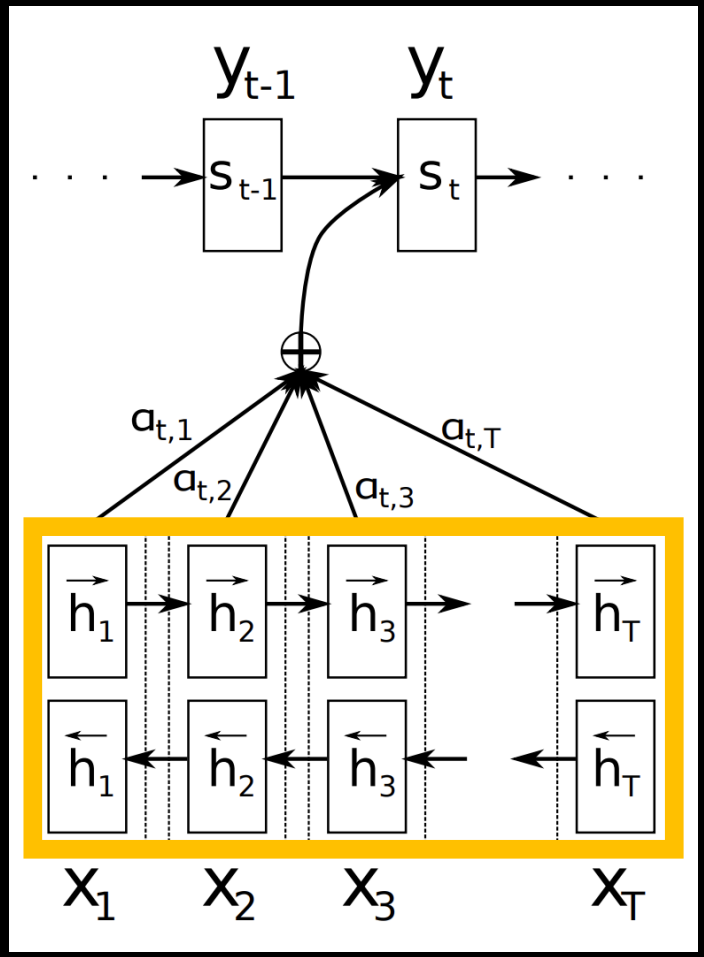
Problem: Attention Uses RNN's Hidden Features, Which are Calculated Sequentially (i.e., Slow)



http://www.wildml.com/2015/09/recurrent-neural-networks-tutorial-part-1-introduction-to-rnns/

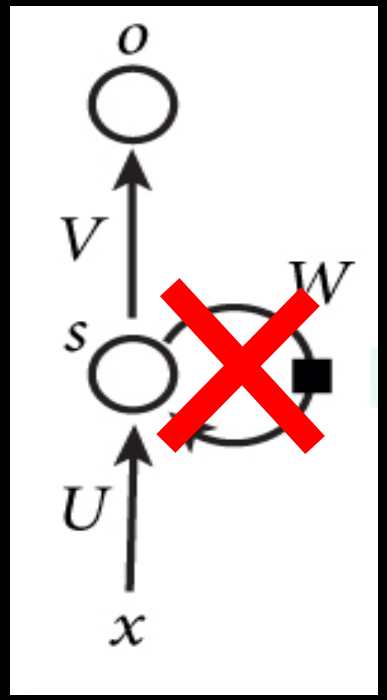
Recurrence is sequential ☹

e.g., RNNSearch model from Bahdanau 2015

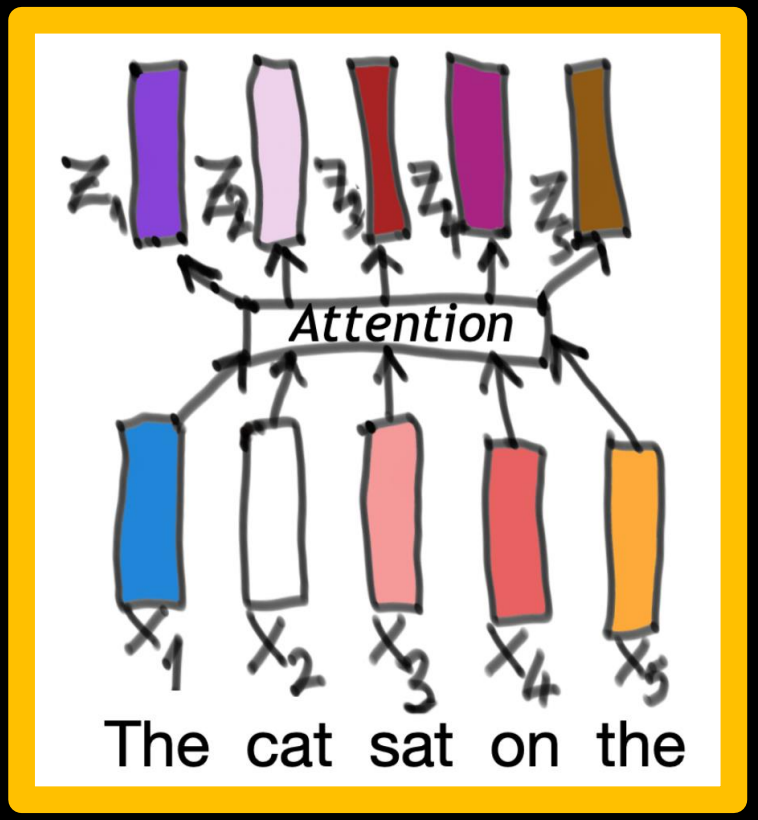


[Bahdanau, Cho, and Bengio]. ICLR 2015]

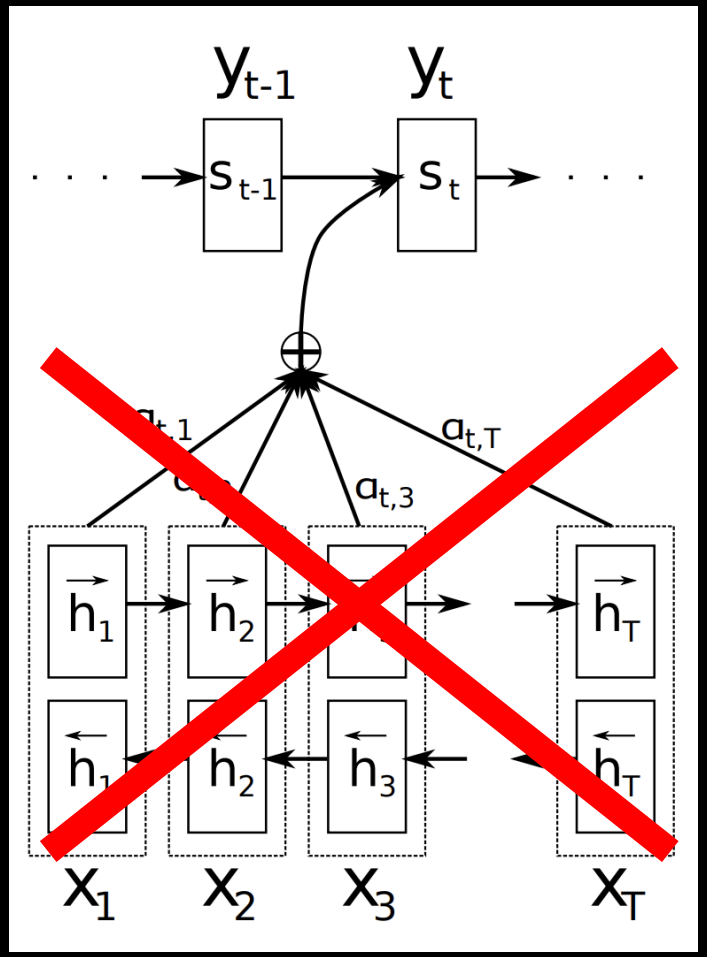
Idea: Replace Recurrence with Parallelizable (Self-)Attention



http://www.wildml.com/2015/09/recurrent-neural-networks-tutorial-part-1-introduction-to-rnns/

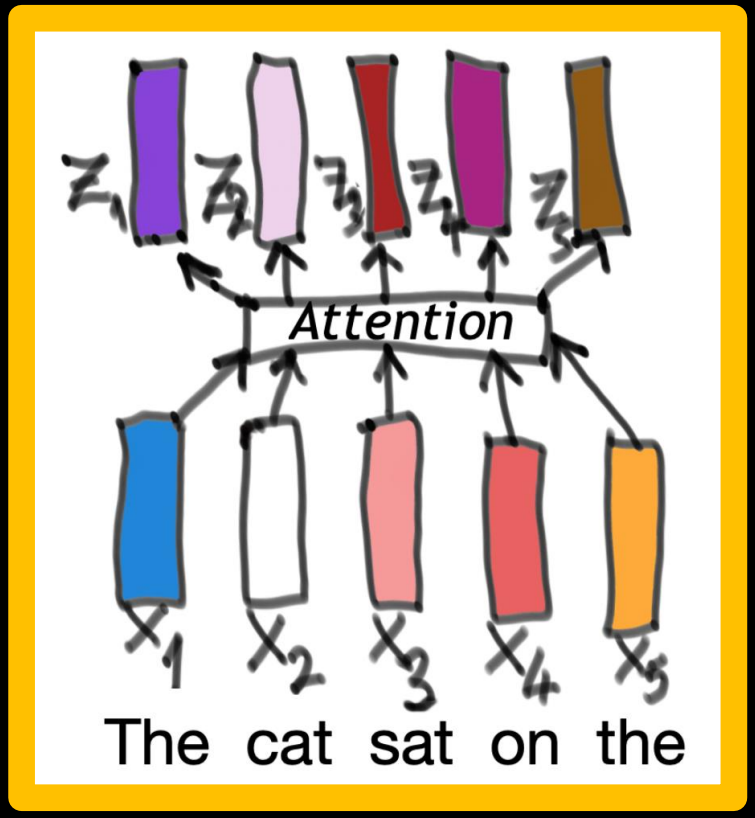


https://towardsdatascience.com/self-attention-5b95ea164f61



[Bahdanau, Cho, and Bengio]. ICLR 2015]

Transformer Architecture: Only Uses Self-Attention to Represent Sequences



https://towardsdatascience.com/self-attention-5b95ea164f61

(NeurIPS, 2017)

Attention Is All You Need

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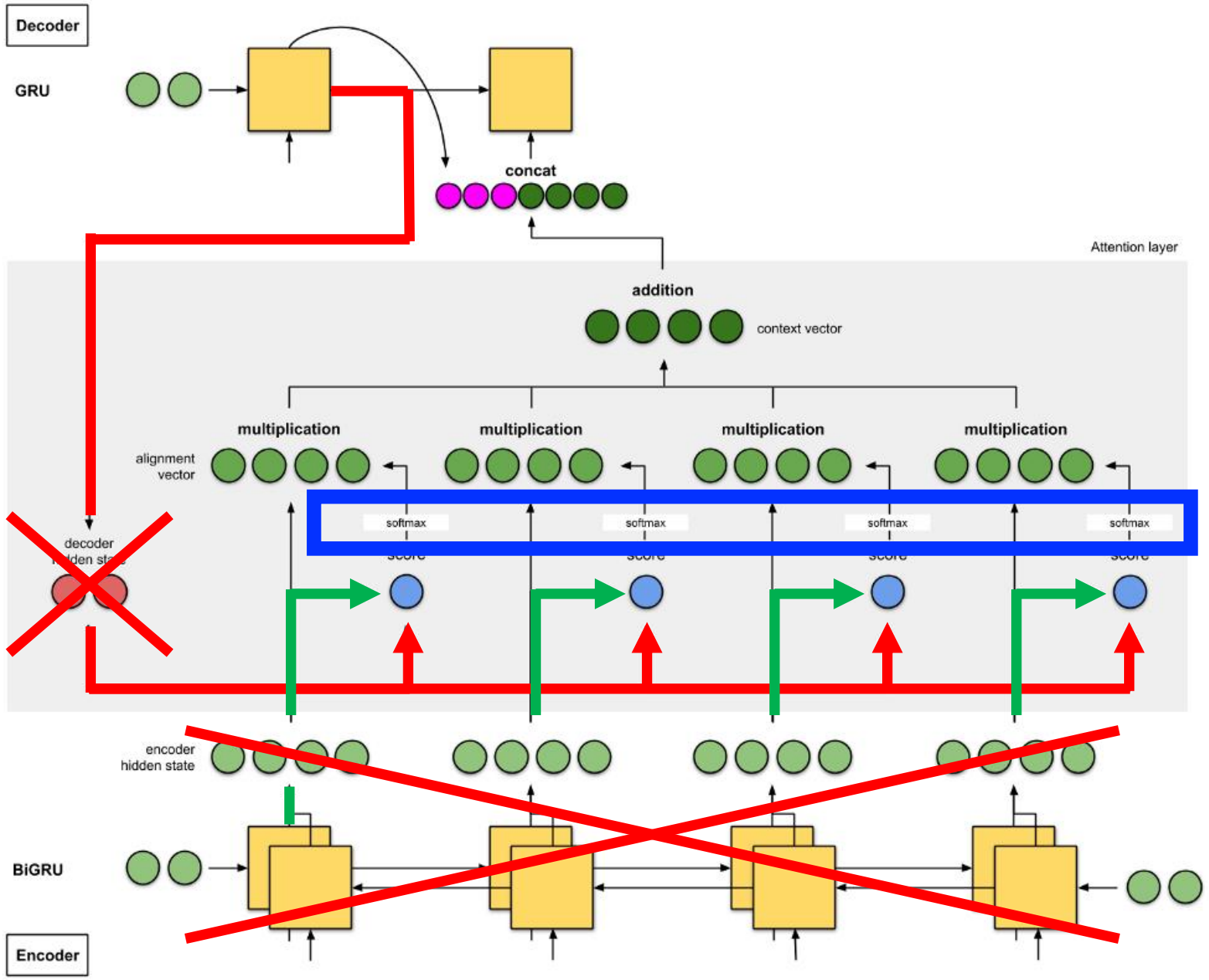
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Context

Recall RNN with Attention:
- indirectly use each input to calculate attention weights, using hidden states from recurrent units

Idea:
- directly use each input to calculate attention weights



https://towardsdatascience.com/attn-illustrated-attention-5ec4ad276ee3

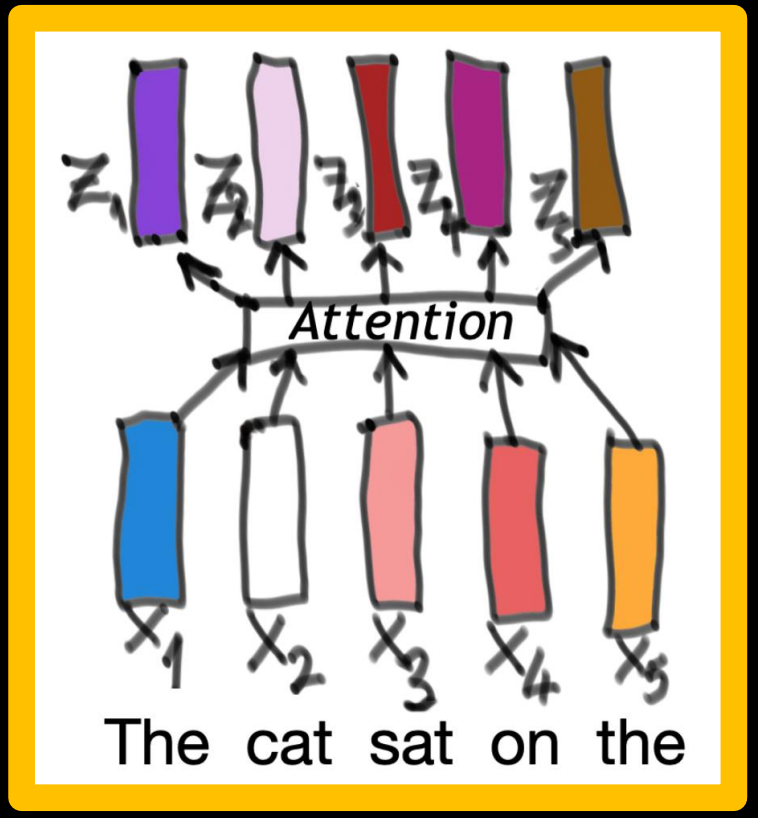
(EMNLP, 2016)

Long Short-Term Memory-Networks for Machine Reading

Jianpeng Cheng, Li Dong and Mirella Lapata

Self-attention: “In our model, memory and attention are added within a sequence encoder allowing the network to uncover lexical relations between tokens.”

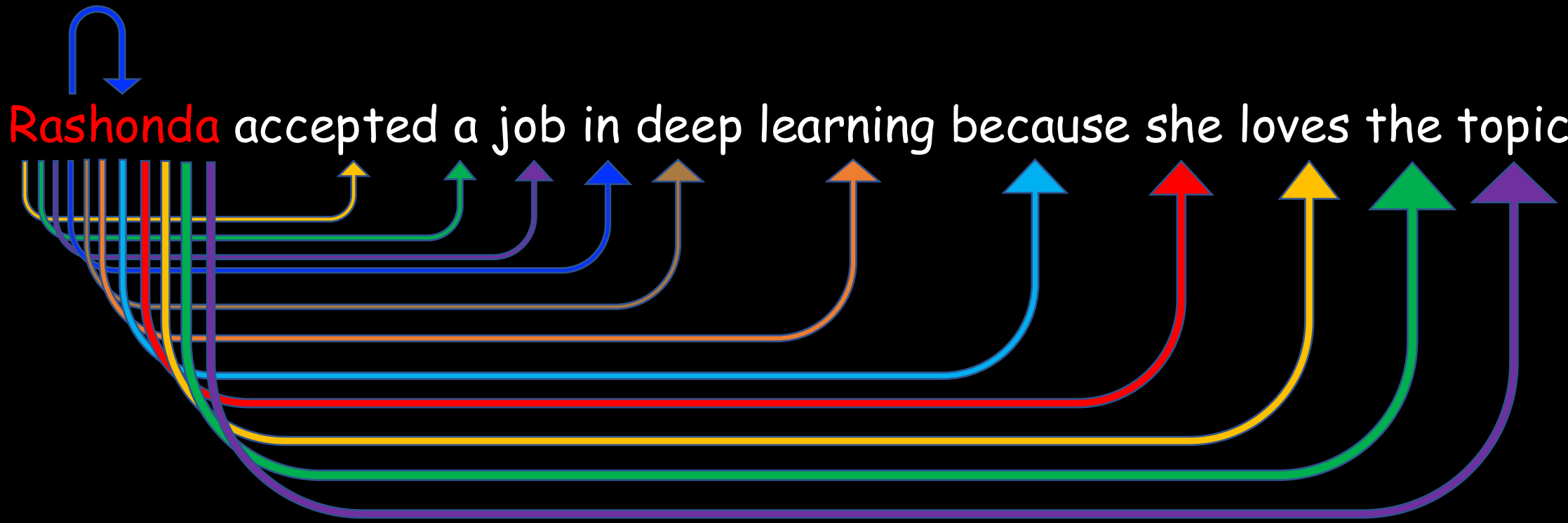
Self-Attention: Every Input Updated Into a Context-aware Representation, *in Parallel*



https://towardsdatascience.com/self-attention-5b95ea164f61

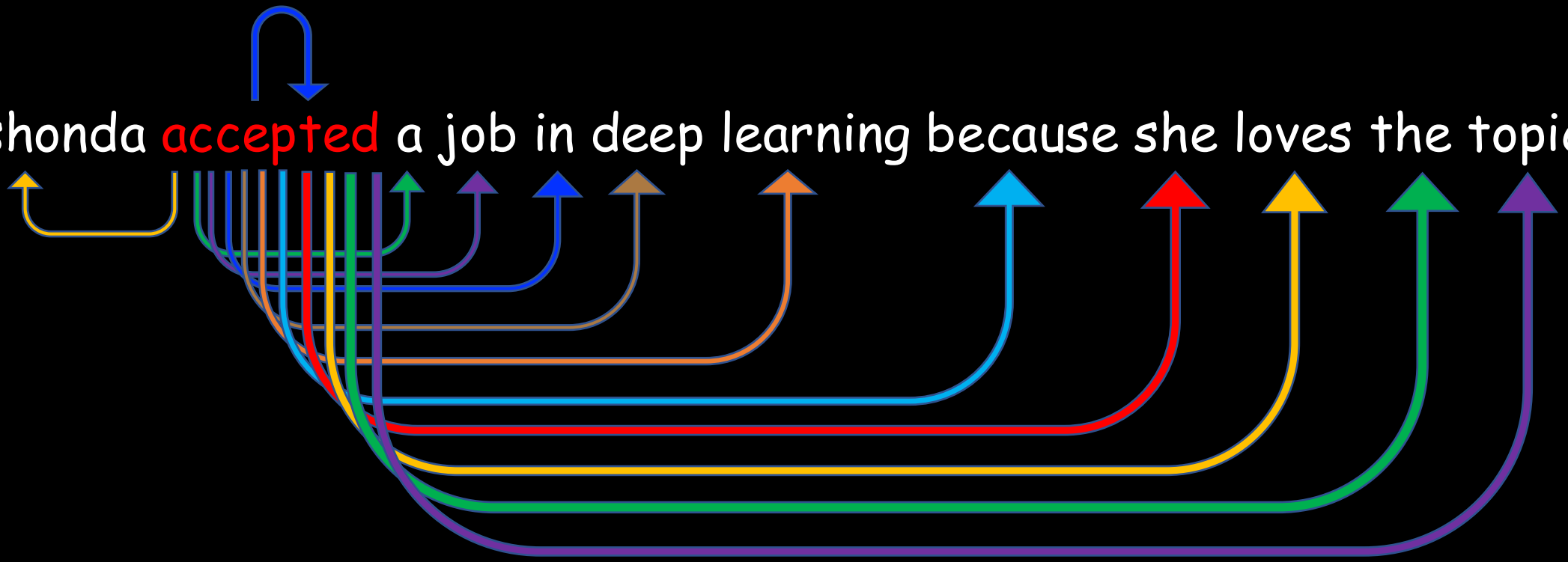
Self-Attention: Every Input Updated Into a Context-aware Representation, *in Parallel*

New representation of each token in a sequence showing its relationship to all tokens; e.g.,



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Self-Attention: Every Input Updated Into a Context-aware Representation, *in Parallel*

New representation of each token in a sequence showing its relationship to all tokens; e.g.,

Rashonda accepted a job in deep learning because she loves the topic



And so on for remaining words…

Self-Attention: Disambiguates Word Meanings

New representation of each token in a sequence showing its relationship to all tokens; e.g.,

Rashonda accepted a job in deep learning because she loves the topic

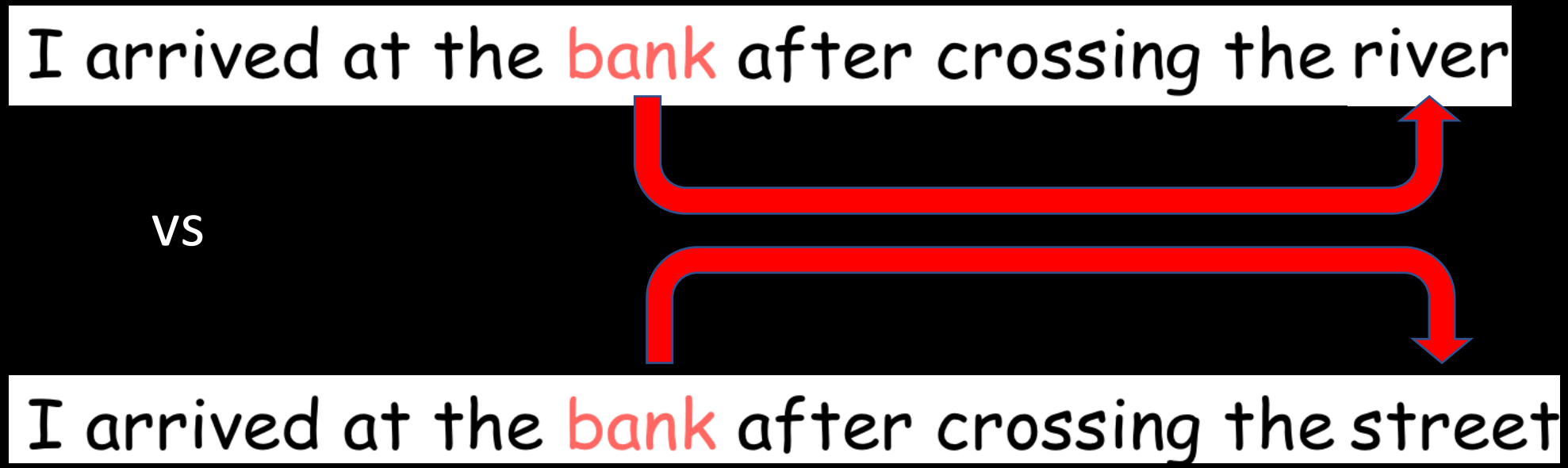


A large attention score means the other word will strongly inform the new representation of the word;
e.g., “she” would encode information about “Rashonda”

Self-Attention: Disambiguates Word Meanings

What does bank mean in this sentence?

- e.g., high attention score with “river” vs with “street disambiguates meaning

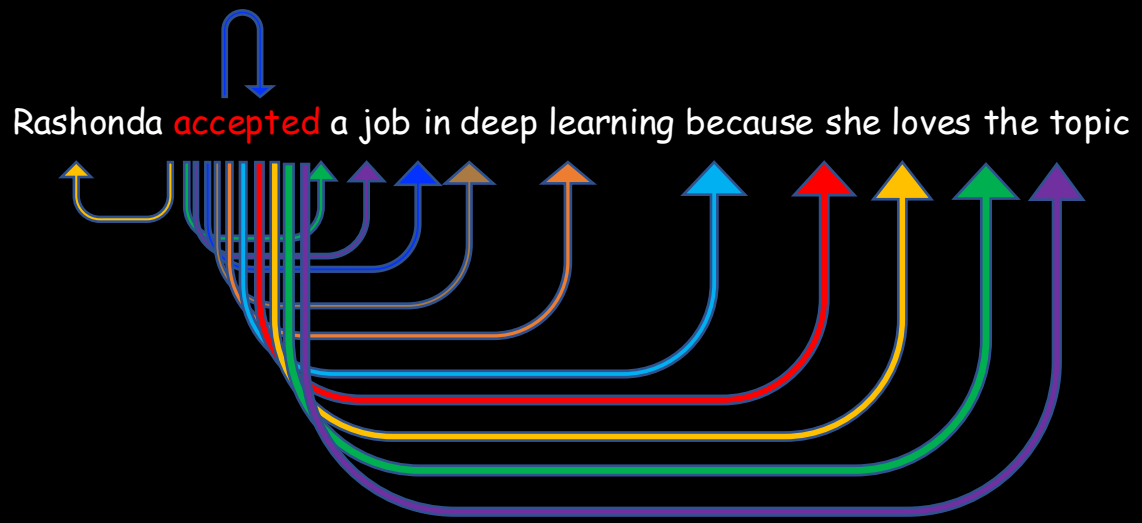


https://lena-voita.github.io/nlp_course/seq2seq_and_attention.html#transformer_intro

Self-Attention vs Cross-Attention: Attends to the Same vs a Different Sequence

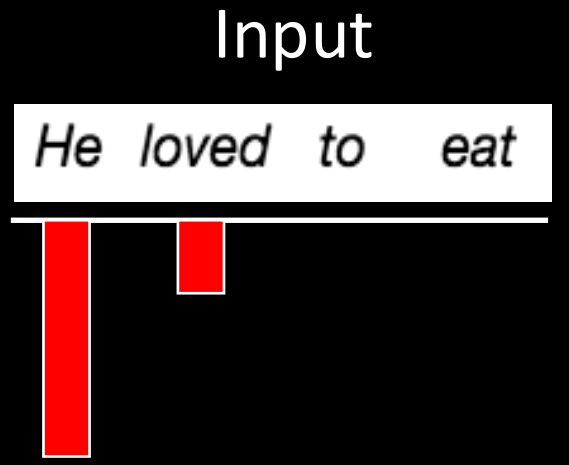
Self-attention

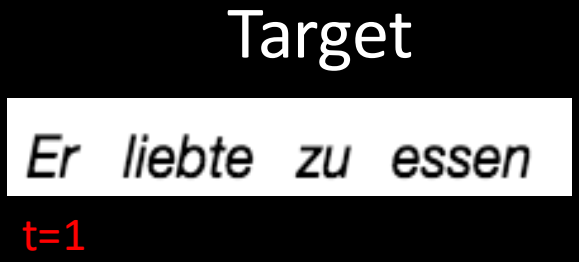
Relates tokens from the same source



Cross-attention

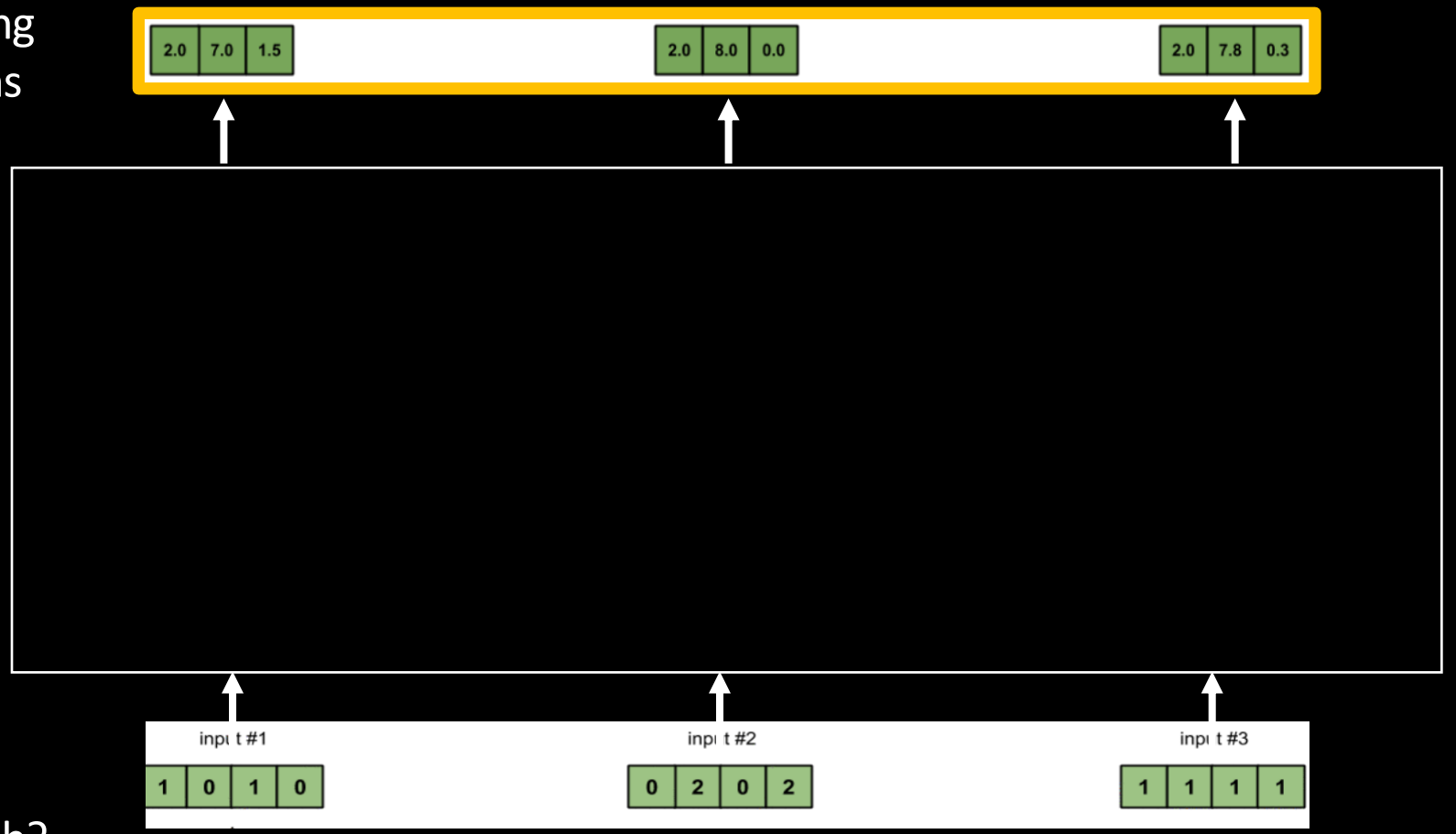
Relates tokens from different sources





Self-Attention: Example

Output: new representations showing each tokens relations to other tokens



Input:

- How many tokens are there?
- What is the dimensionality of each?

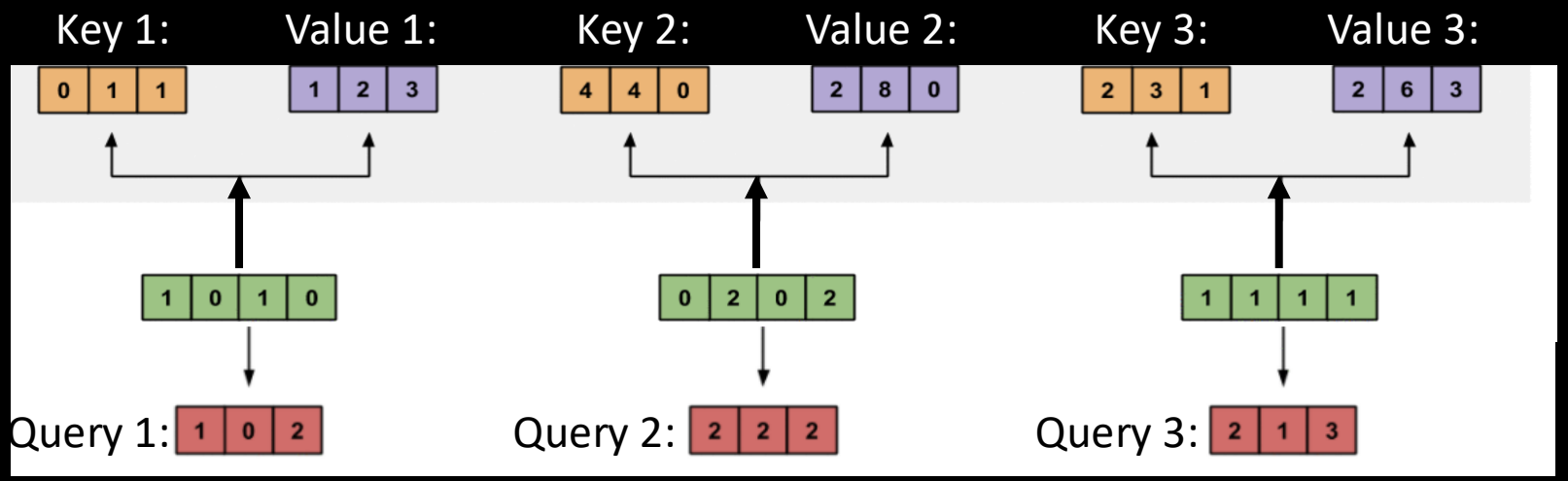
https://towardsdatascience.com/illustrated-self-attention-2d627e33b20a

Self-Attention: Example

For each input:

- Compute attention weight from query and key vectors
- Compute new representation using attention weights and value vector

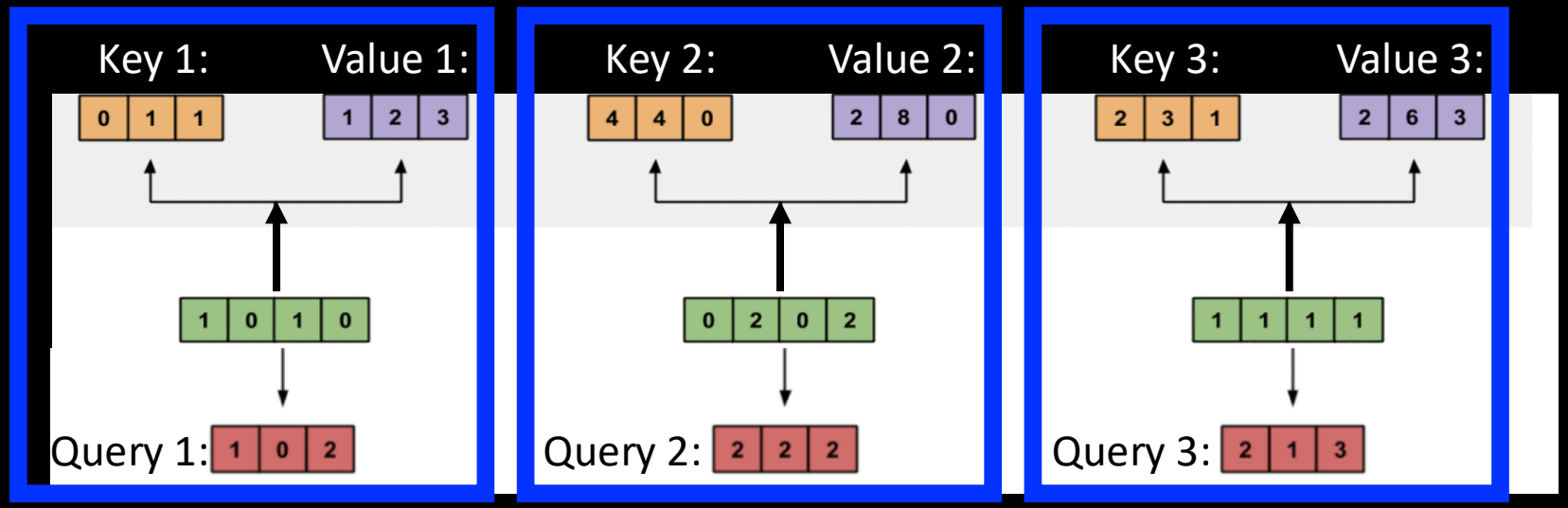
Let's go through the steps…



https://towardsdatascience.com/illustrated-self-attention-2d627e33b20a

Self-Attention: Example

Each input is multiplied by 3 weight matrices to create 3 vectors: query, key, and value

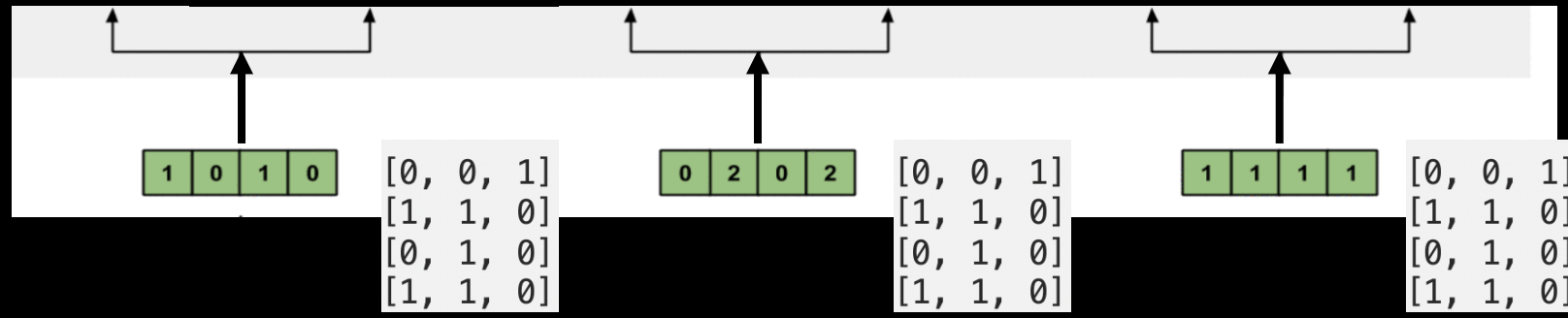


https://towardsdatascience.com/illustrated-self-attention-2d627e33b20a

Self-Attention: Example

e.g., key weights

```
[0, 0, 1]
[1, 1, 0]
[0, 1, 0]
[1, 1, 0]
```

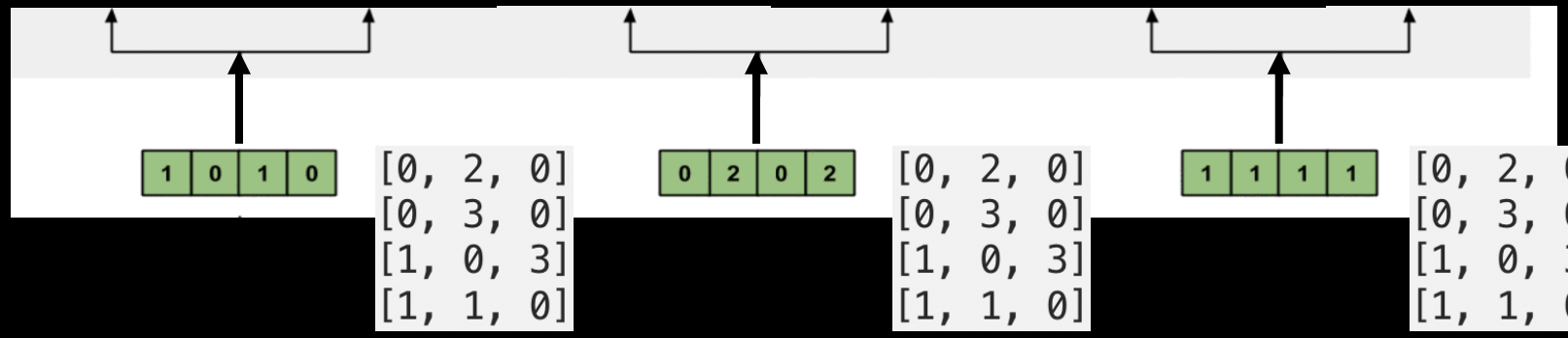


https://towardsdatascience.com/illustrated-self-attention-2d627e33b20a

Self-Attention: Example

e.g., value weights

```
[0, 2, 0]
[0, 3, 0]
[1, 0, 3]
[1, 1, 0]
```

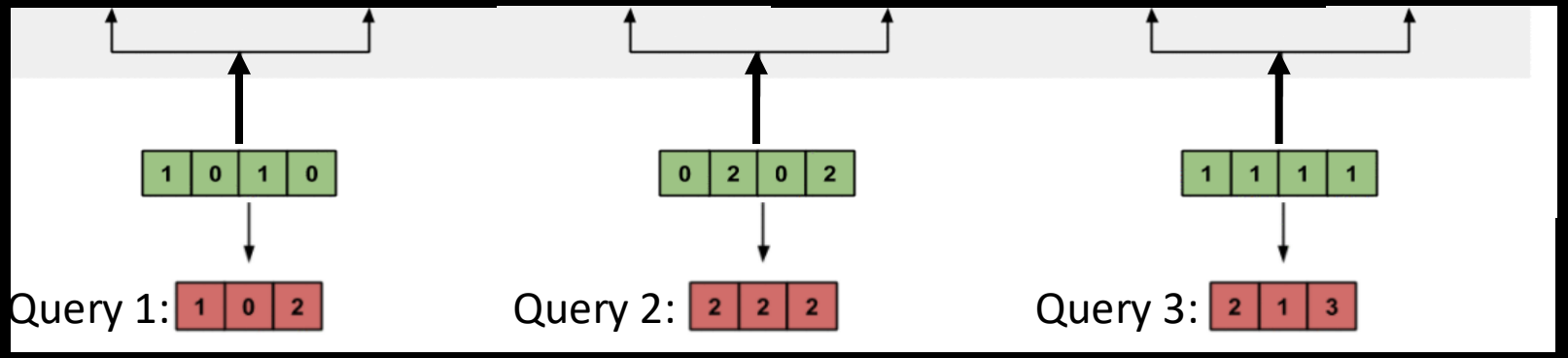


https://towardsdatascience.com/illustrated-self-attention-2d627e33b20a

Self-Attention: Example

e.g., query weights

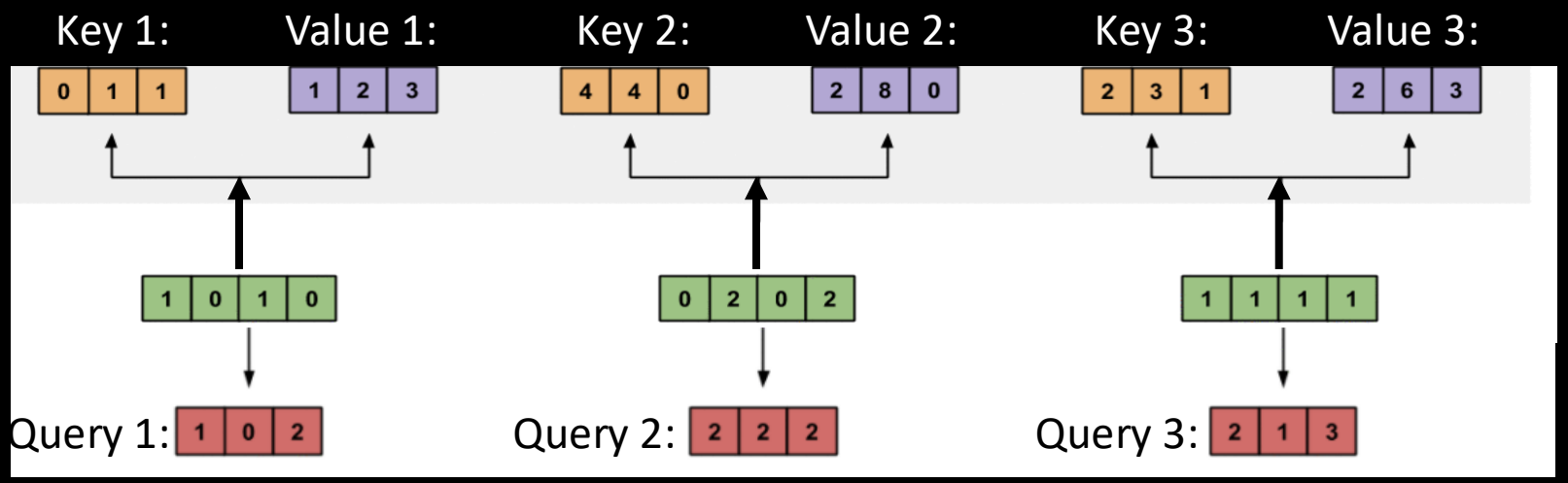
```
[1, 0, 1]
[1, 0, 0]
[0, 0, 1]
[0, 1, 1]
```



https://towardsdatascience.com/illustrated-self-attention-2d627e33b20a

Self-Attention: Example

How many weight matrices are learned in this example?



https://towardsdatascience.com/illustrated-self-attention-2d627e33b20a

Self-Attention: Example (Input 1 Update)

e.g., dot product

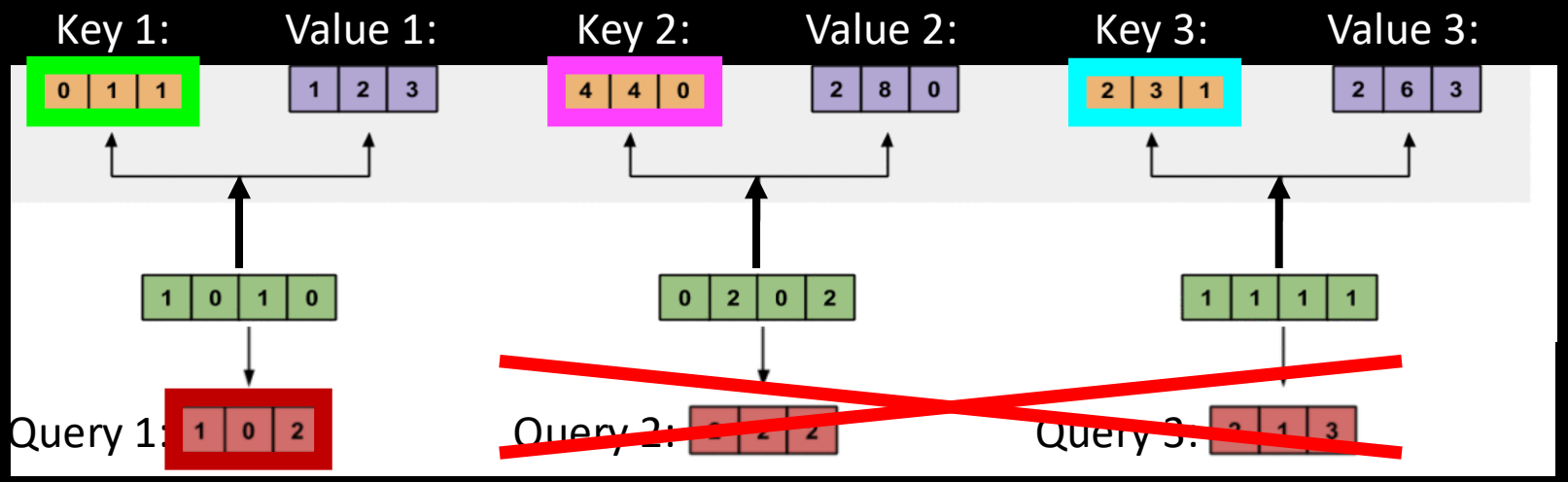
1	0	2	X	0	= 2
				1	
				1	

1	0	2	X	4	= 4
				4	
				0	

1	0	2	X	2	= 4
				3	
				1	

Query ("What I am looking for")

Keys ("What I have")



https://towardsdatascience.com/illustrated-self-attention-2d627e33b20a

Self-Attention: Example (Input 1 Update)

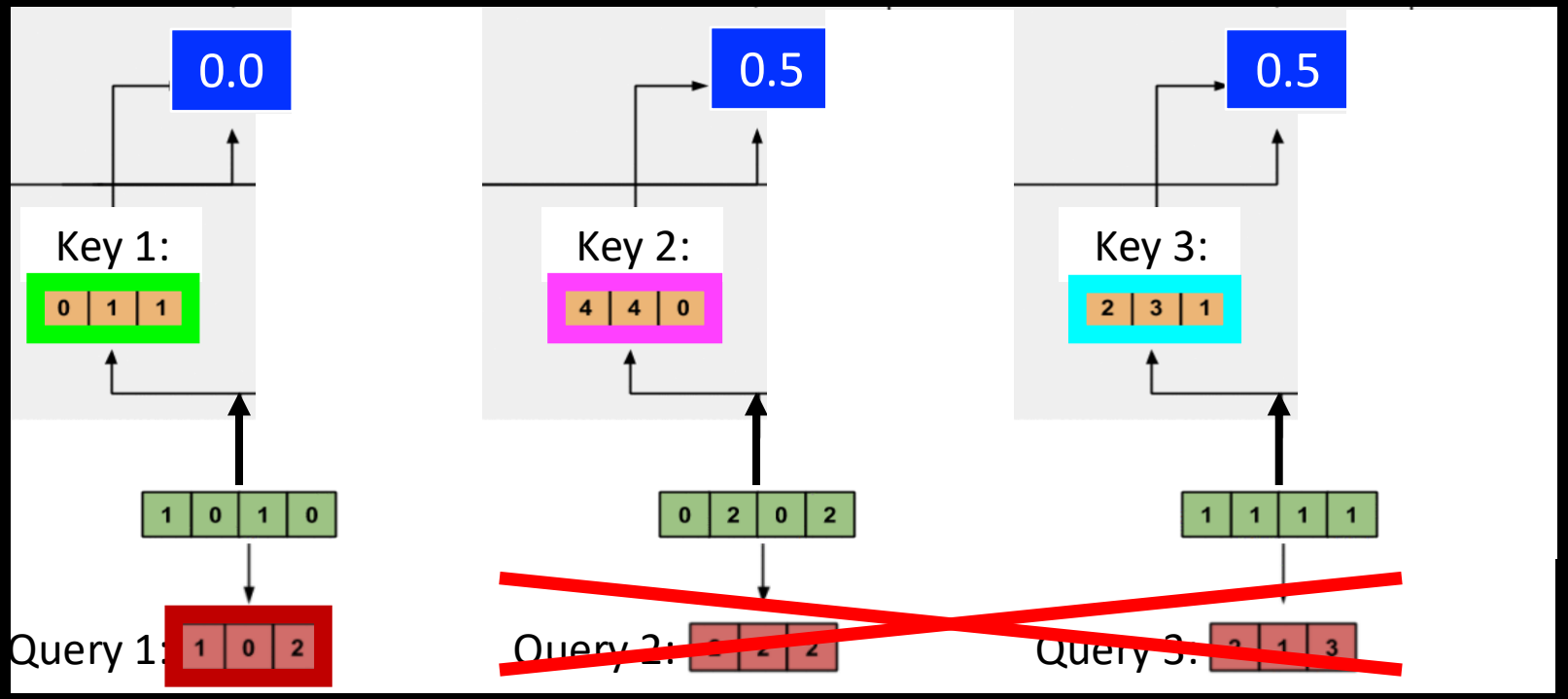
e.g., dot product values passed through softmax for attention weights

= softmax([2, 4, 4])

= [0.0, 0.5, 0.5])

How can 0 arise from a softmax function?
- rounding

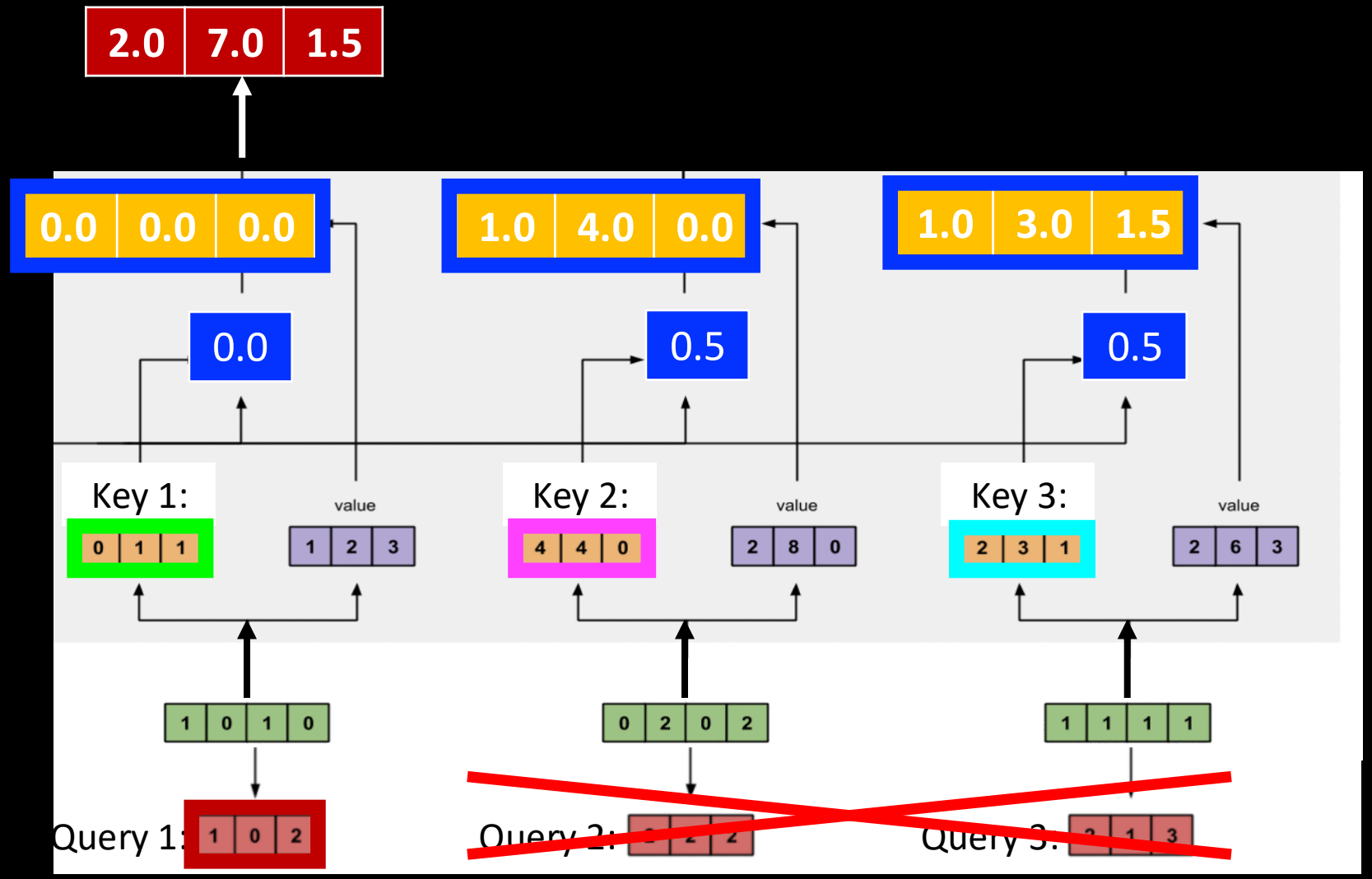
To which input(s) will input 1 be:
- least related?
- most related?



https://towardsdatascience.com/illustrated-self-attention-2d627e33b20a

Self-Attention: Example (Input 1 Update)

e.g., dot product values passed through softmax for attention weights, and then used to compute weighted sum of value vectors (”what I contribute”) for new representation



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Self-Attention: Example (Input 2 Update)

e.g., dot product

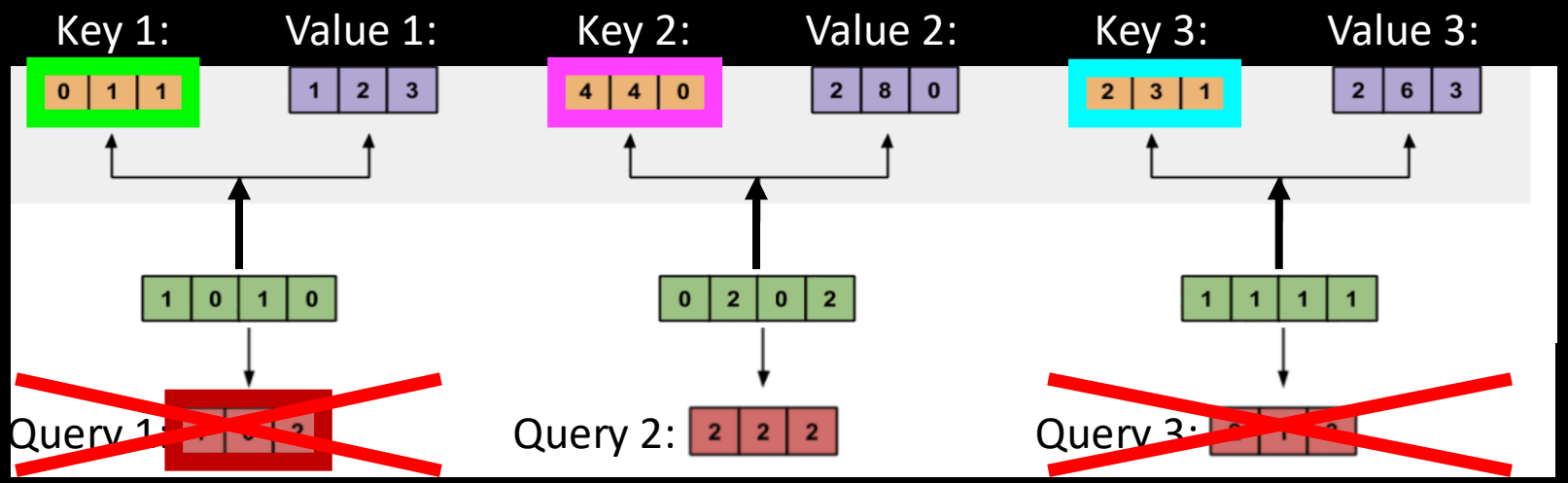
2	2	2	X	0	1	1	= 4

2	2	2	X	4	4	0	= 16

2	2	2	X	2	3	1	= 12

Query ("What I am looking for")

Keys ("What I have")



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Self-Attention: Example (Input 2 Update)

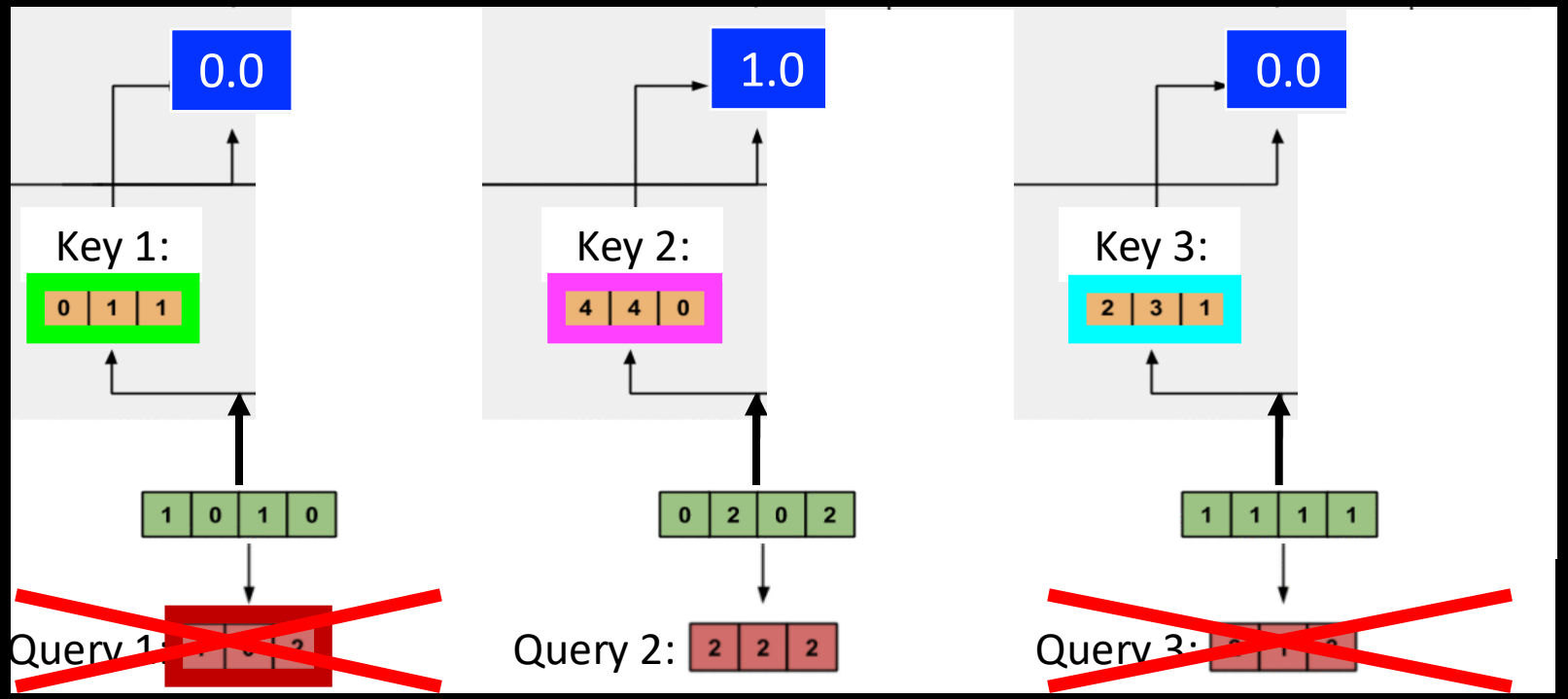
e.g., dot product values passed through softmax for attention weights

= softmax([4, 16, 12])

= [0.0, 1.0, 0.0])

Recall how 0 can arise from a softmax function:
- rounding

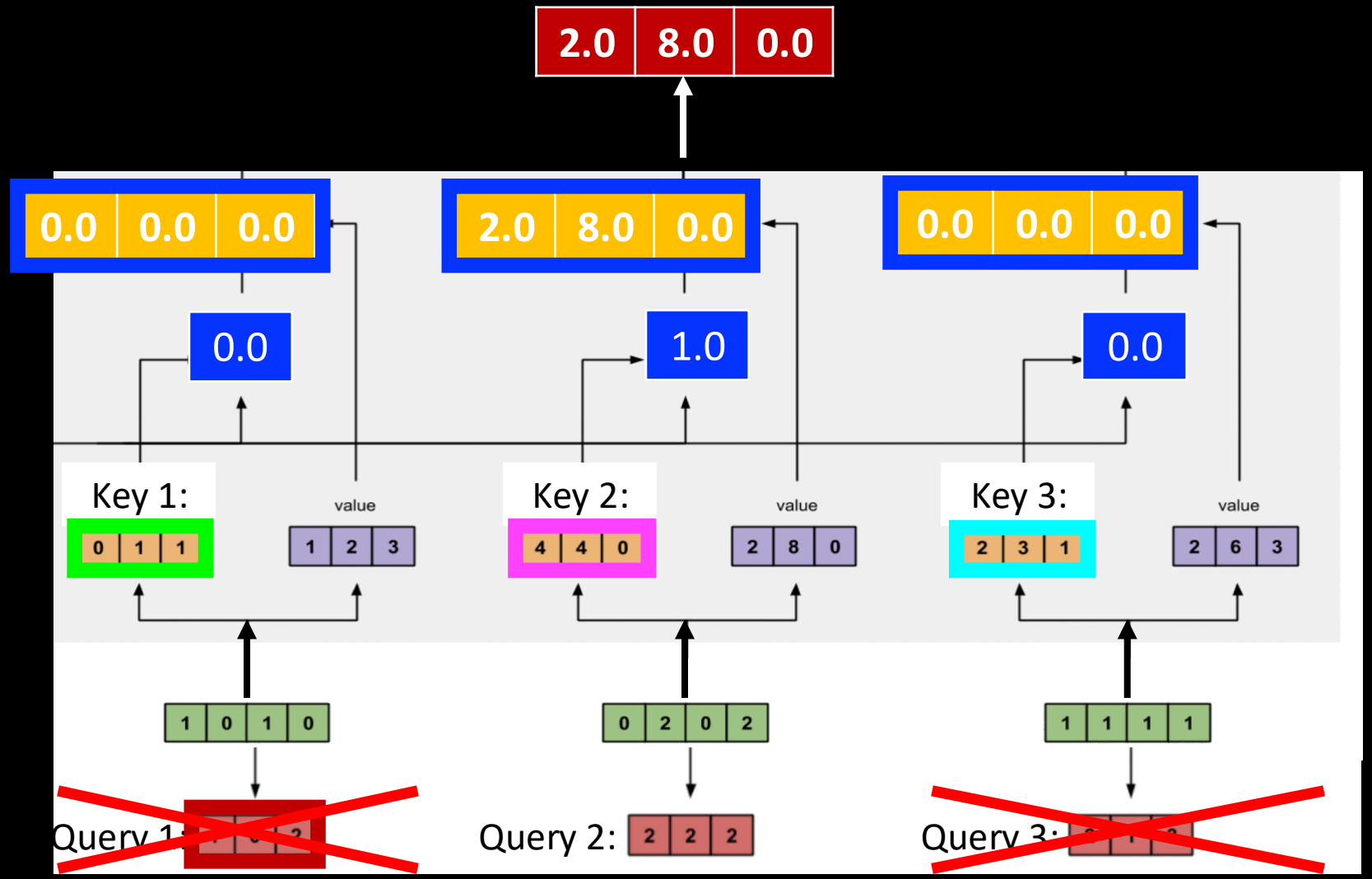
To which input(s) will input 2 be:
- least related?
- most related?



https://towardsdatascience.com/illustrated-self-attention-2d627e33b20a

Self-Attention: Example (Input 2 Update)

e.g., dot product values passed through softmax for attention weights, and then used to compute weighted sum of value vectors (”what I contribute”) for new representation



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Self-Attention: Example (Input 3 Update)

e.g., dot product

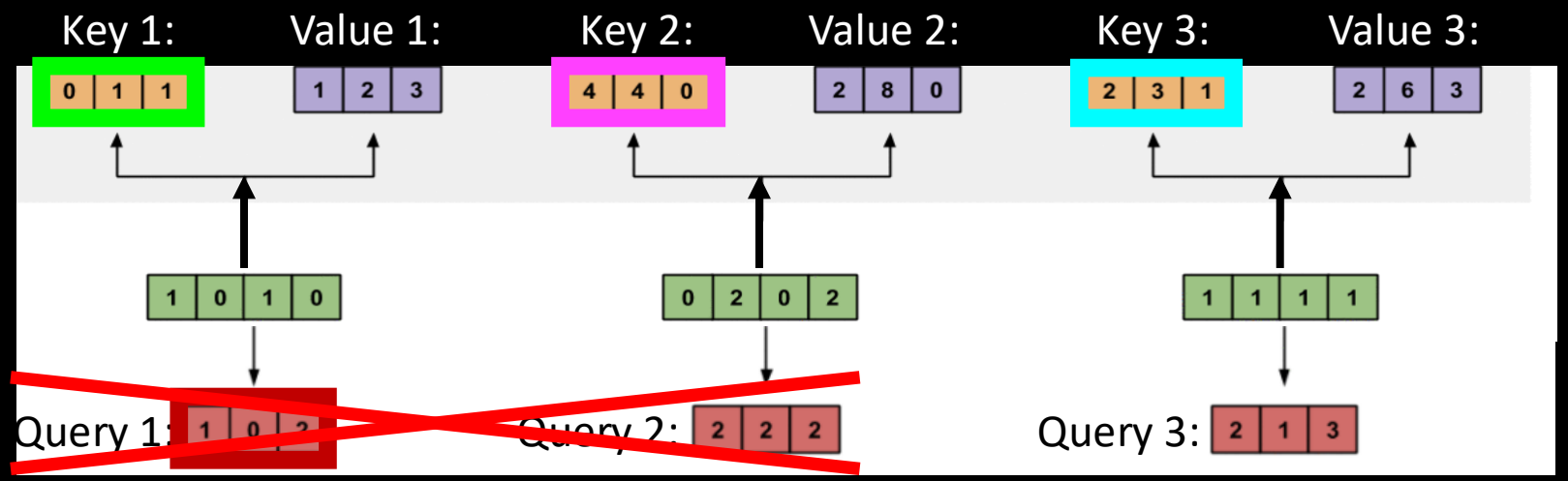
2	1	3	X	0	1	1	= 4

2	1	3	X	4	4	0	= 12

2	1	3	X	2	3	1	= 10

Query ("What I am looking for")

Keys ("What I have")



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Self-Attention: Example (Input 3 Update)

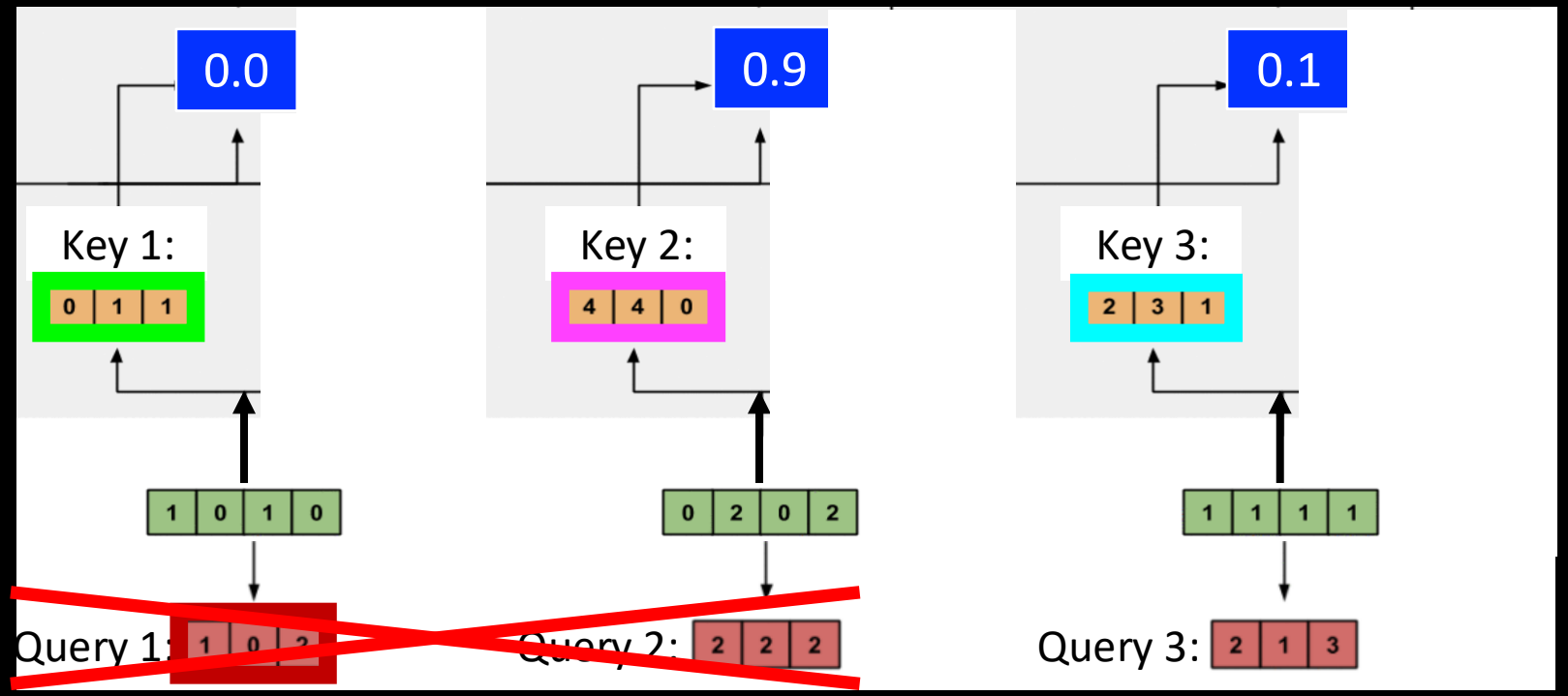
e.g., dot product values passed through softmax for attention weights

= softmax([4, 12, 10])

= [0.0, 0.9, 0.1])

Recall how 0 can arise from a softmax function:
- rounding

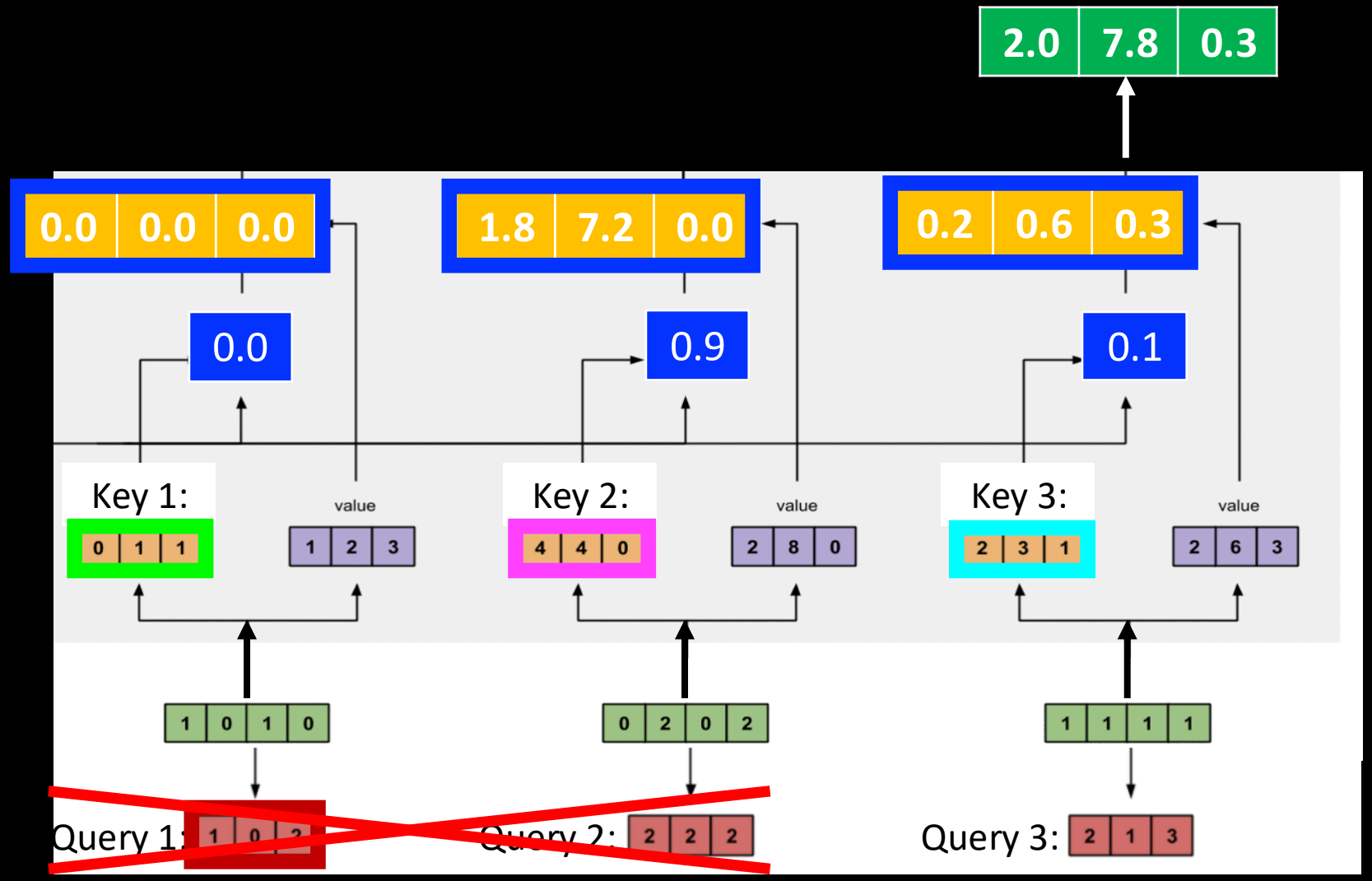
To which input(s) will input 3 be:
- least related?
- most related?



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Self-Attention: Example (Input 3 Update)

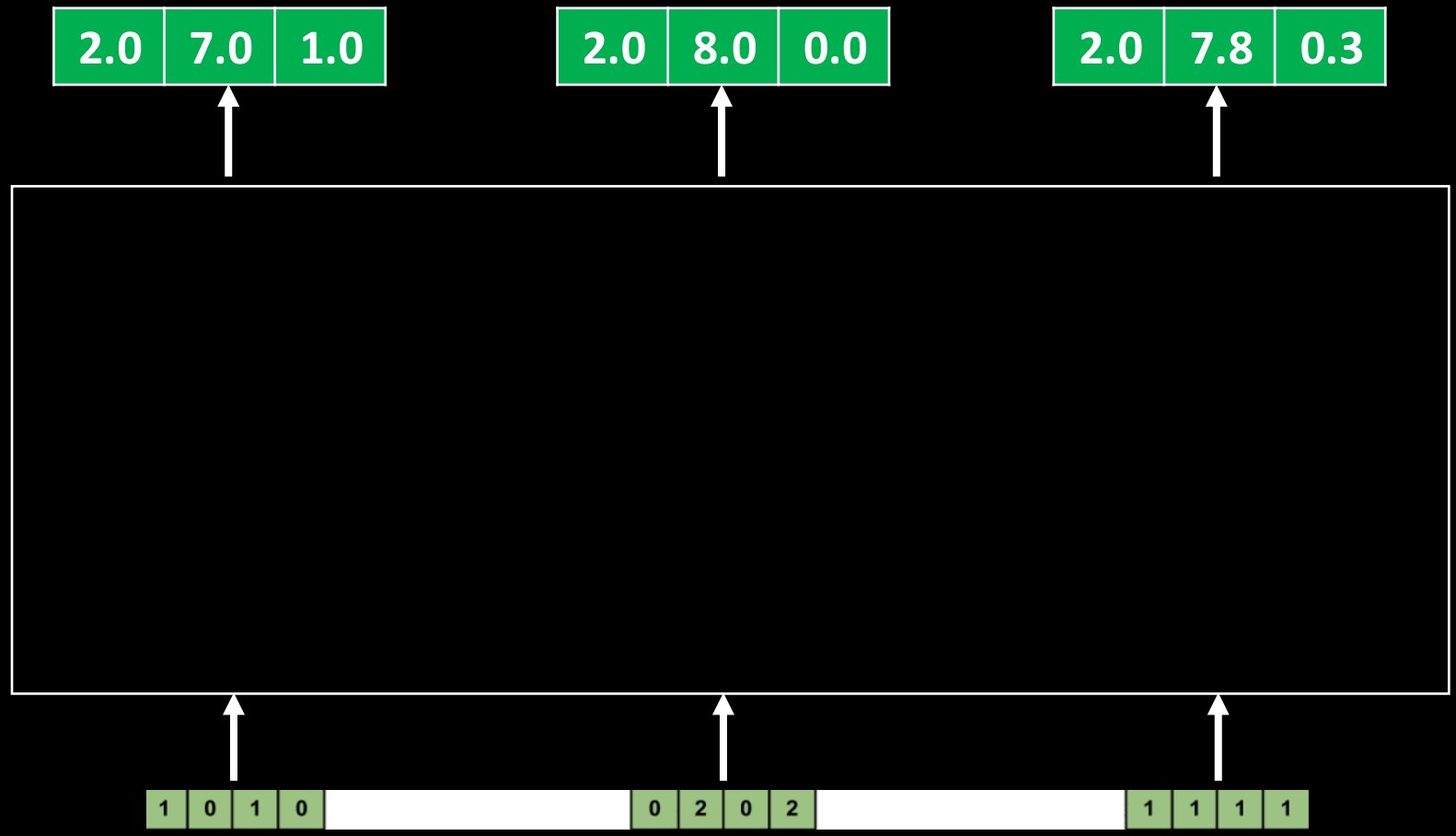
e.g., dot product values passed through softmax for attention weights, and then used to compute weighted sum of value vectors (”what I contribute”) for new representation



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Hyperparameters

- Input token length
- Number of matrices' columns (must match for query and key)

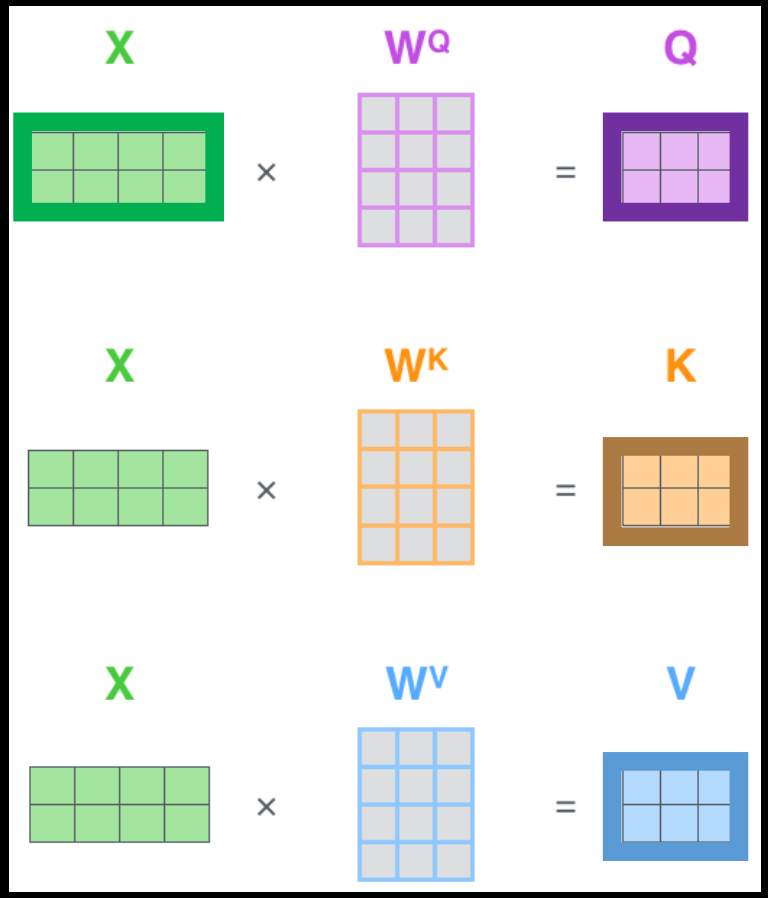


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Self-Attention: Efficient Computation

Step 1

Each row is an input token:



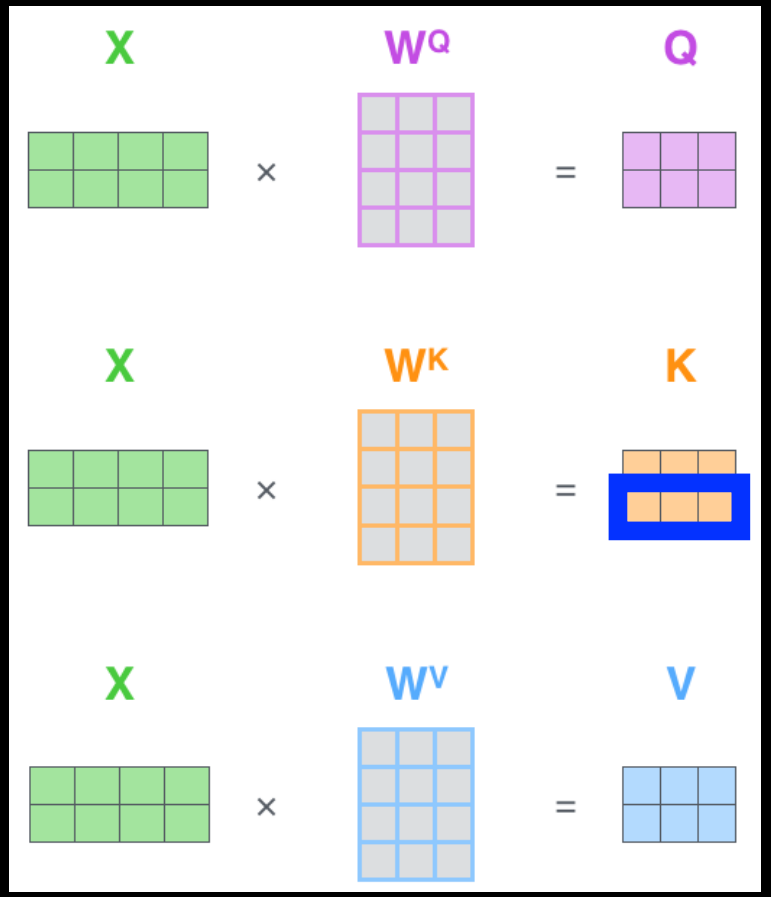
Each row is a query

Each row is a key

Each row is a value

Self-Attention: Efficient Computation

Step 1



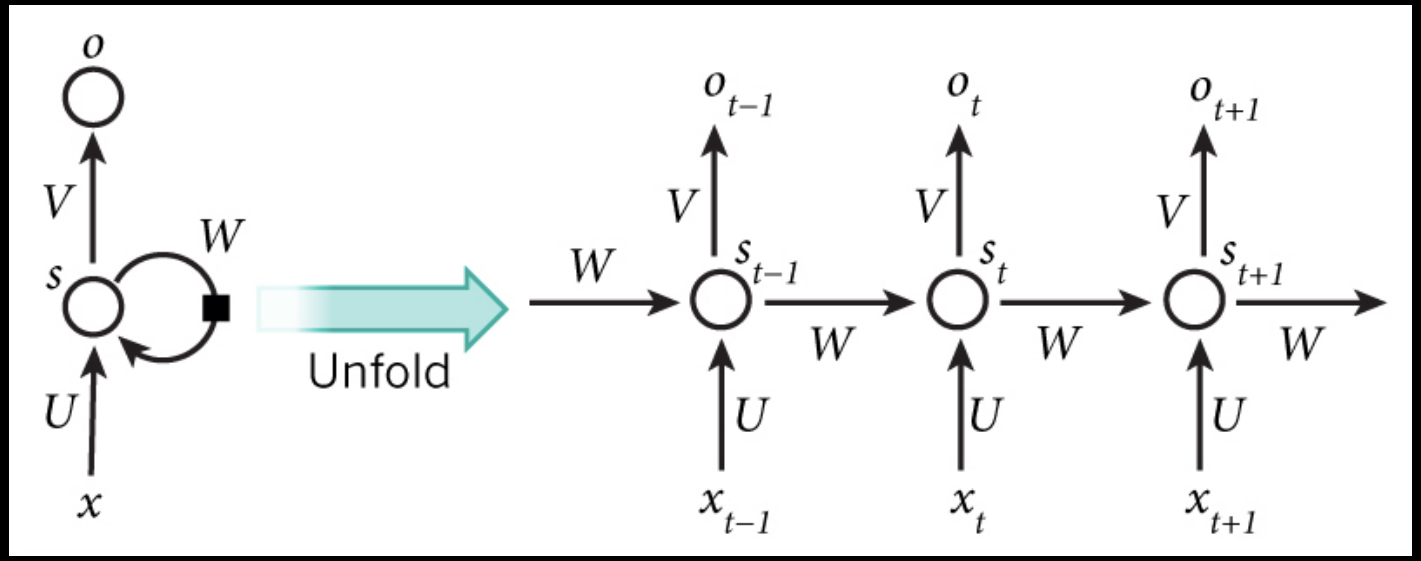
Step 2

$$\text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V = Z$$

Implementation detail: scaling down helps preserve gradients in training; what is the key dimensionality in this example?

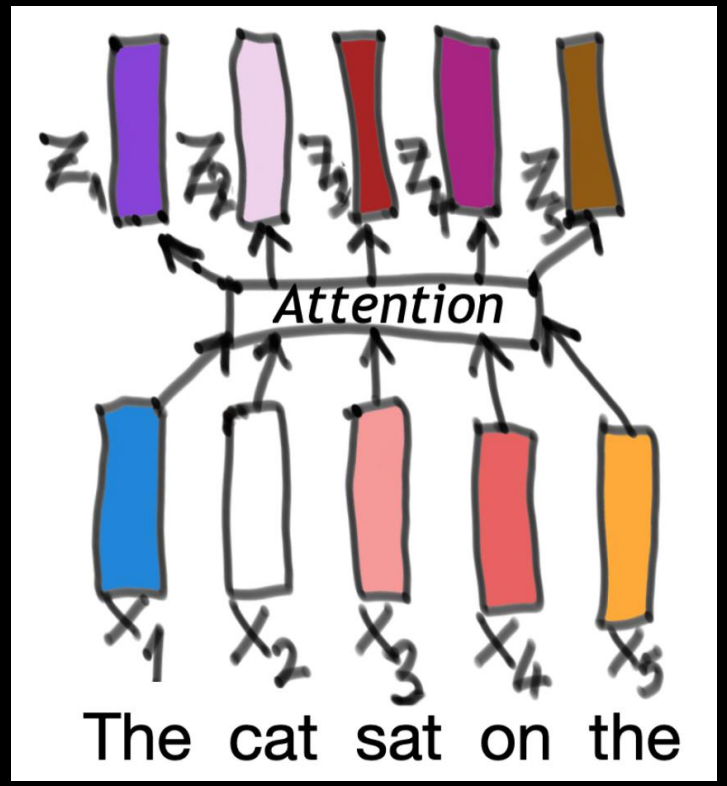
Sequence Representation with Recurrent Unit vs Self-Attention

Sequential Computation



http://www.wildml.com/2015/09/recurrent-neural-networks-tutorial-part-1-introduction-to-rnns/

Parallelized Computation



https://towardsdatascience.com/self-attention-5b95ea164f61

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- Key ingredient: self-attention
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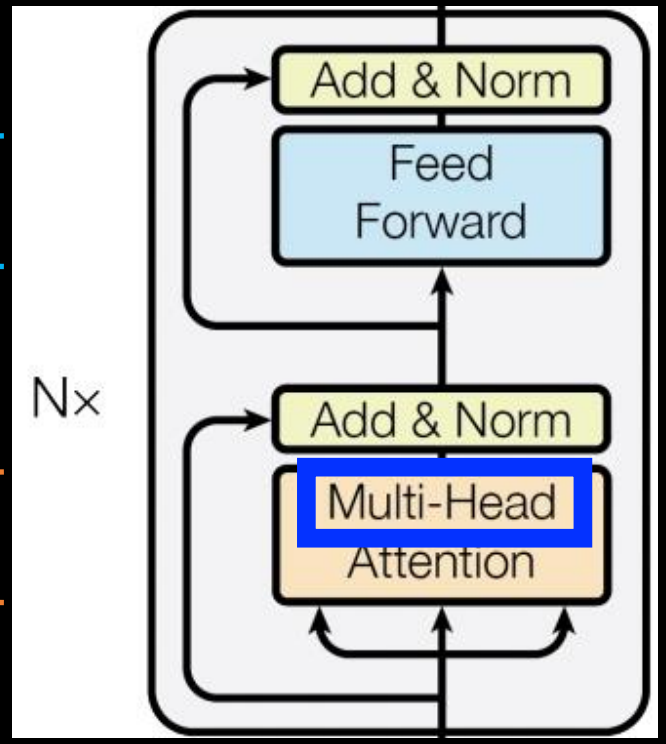
Typical Transformer Block: Stacked Repeatedly

Private reflection: each input updates its views without communication, using a position-wise fully connected layer applied to each separately



Group discussion: all communicate, revising their views based on others





Why include residual connections and layer normalization?

- Improves gradients and stabilizes learning
- Enables ignoring effects of extra attention and feedforward steps (residual connections)

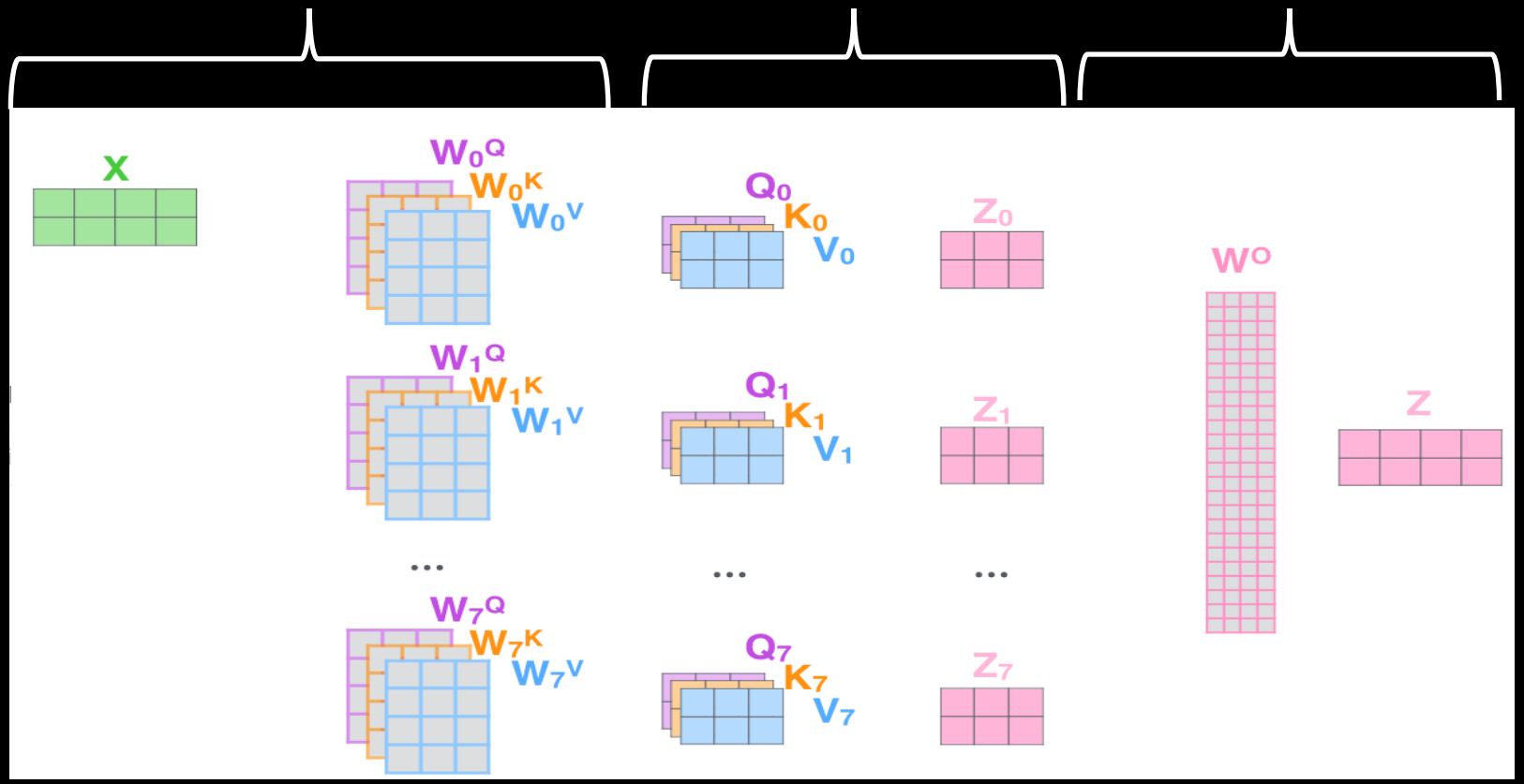
Vaswani et al. Attention Is All You Need. Neurips 2017

Multi-Head Attention

1) Multiple self-attention mechanisms, each with query, key, and value matrices

2) Compute new representations

3) Concatenate new representations (**z_i**) and multiply with weight matrix (w_o) to create a vector size expected by the FF layer



How many attention heads are in this example?

http://jalammar.github.io/illustrated-transformer/

Multi-Head Attention – Examples

Attention weights for first two attention heads

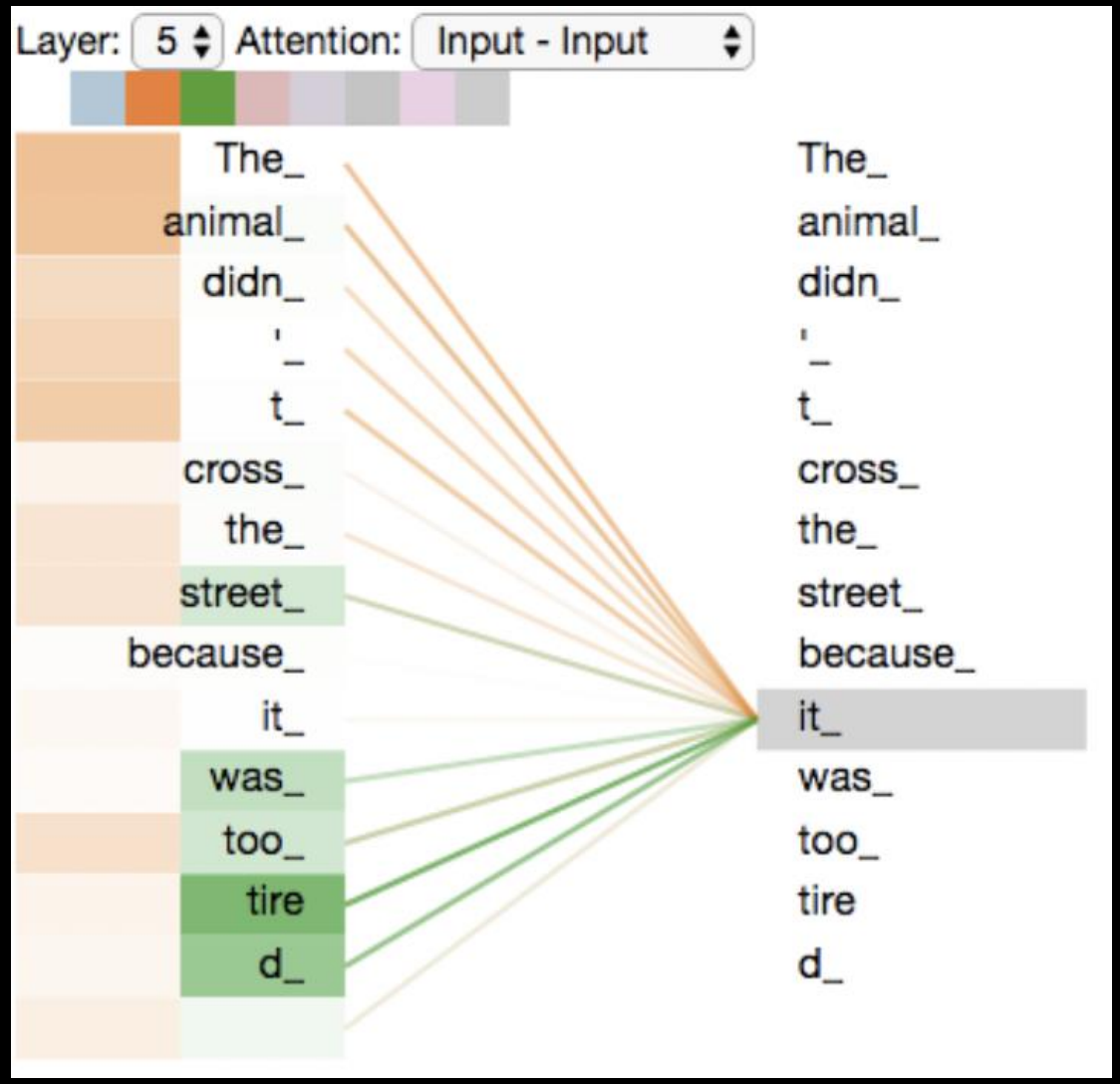
- Darker values signify larger attention scores

What does “it” focus on most in head 1?

- The animal (e.g., represents what is “it”)

What does “it” focus on most in head 2?

- tired (e.g., represents how “it” feels); note, a tokenizer was used that separates “tire” and “d”



Head 1 Head 2

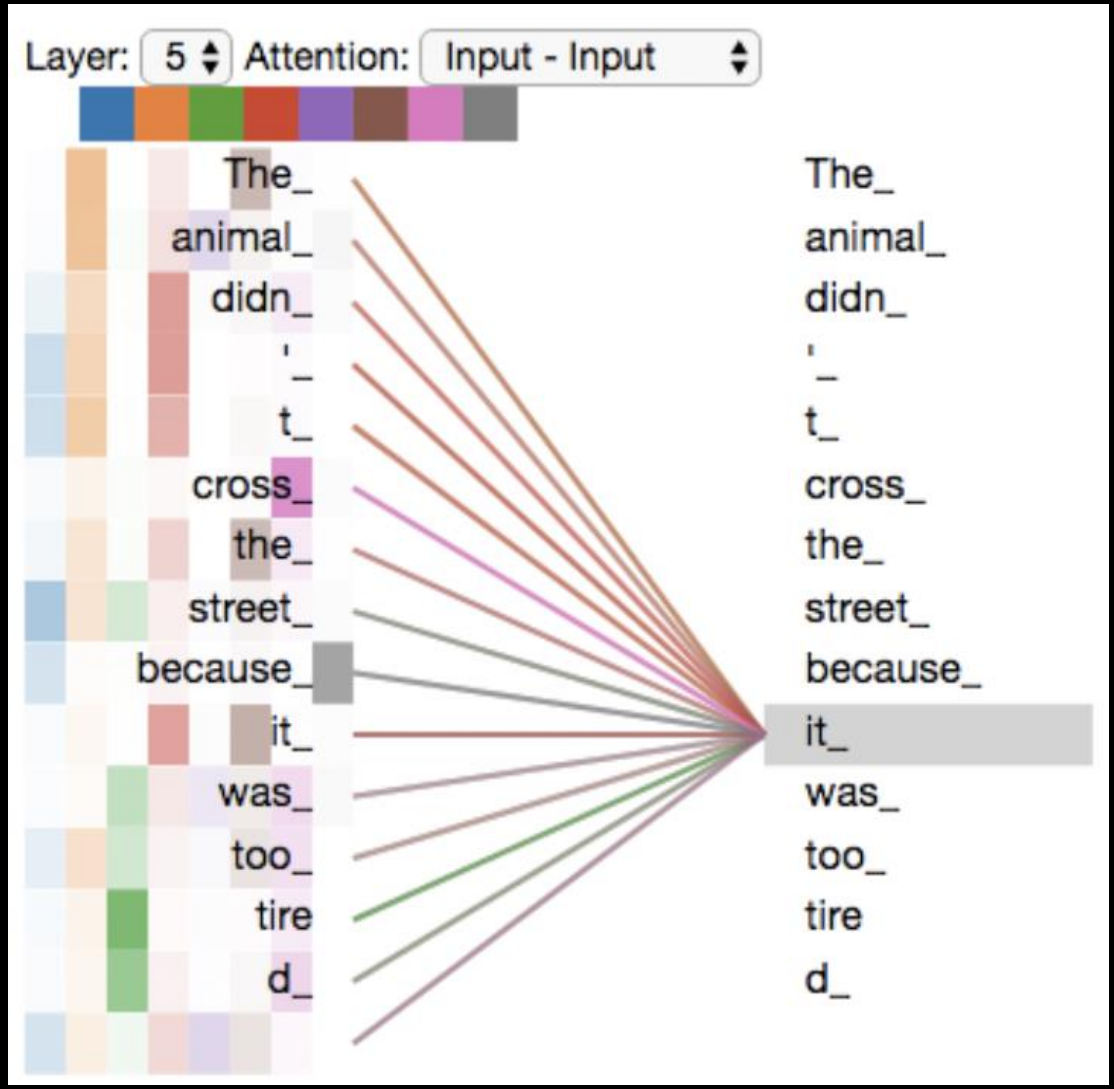
http://jalammar.github.io/illustrated-transformer/

Multi-Head Attention – Examples

Figure shows five columns of attention weights for five attention heads

- Darker values signify larger attention scores

Attention weights may be hard to interpret

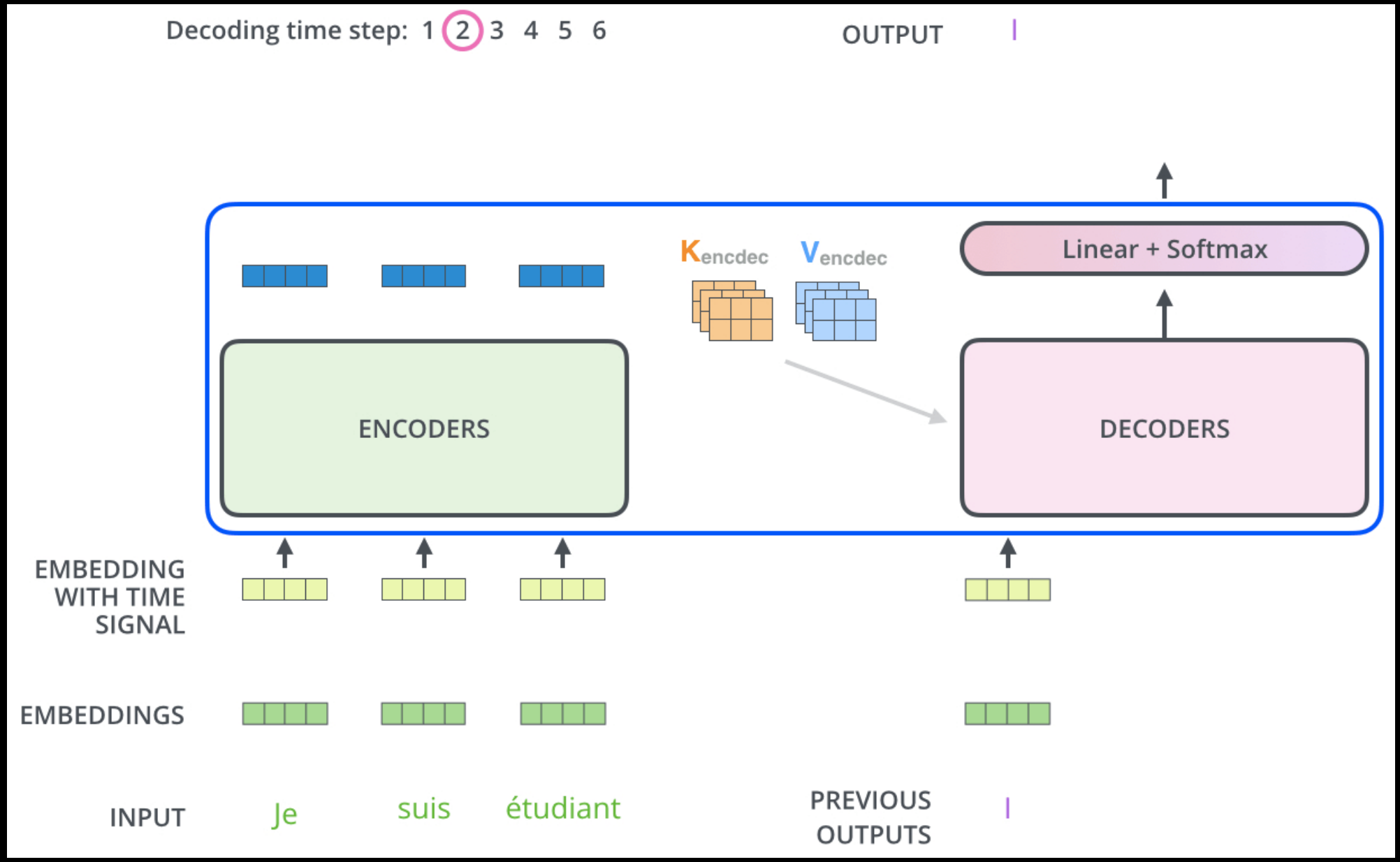


http://jalammar.github.io/illustrated-transformer/

Today’s Topics

- Attention transformed: birth of a new architecture
- Key ingredient: self-attention
- Building a transformer
- Pioneering model for machine translation

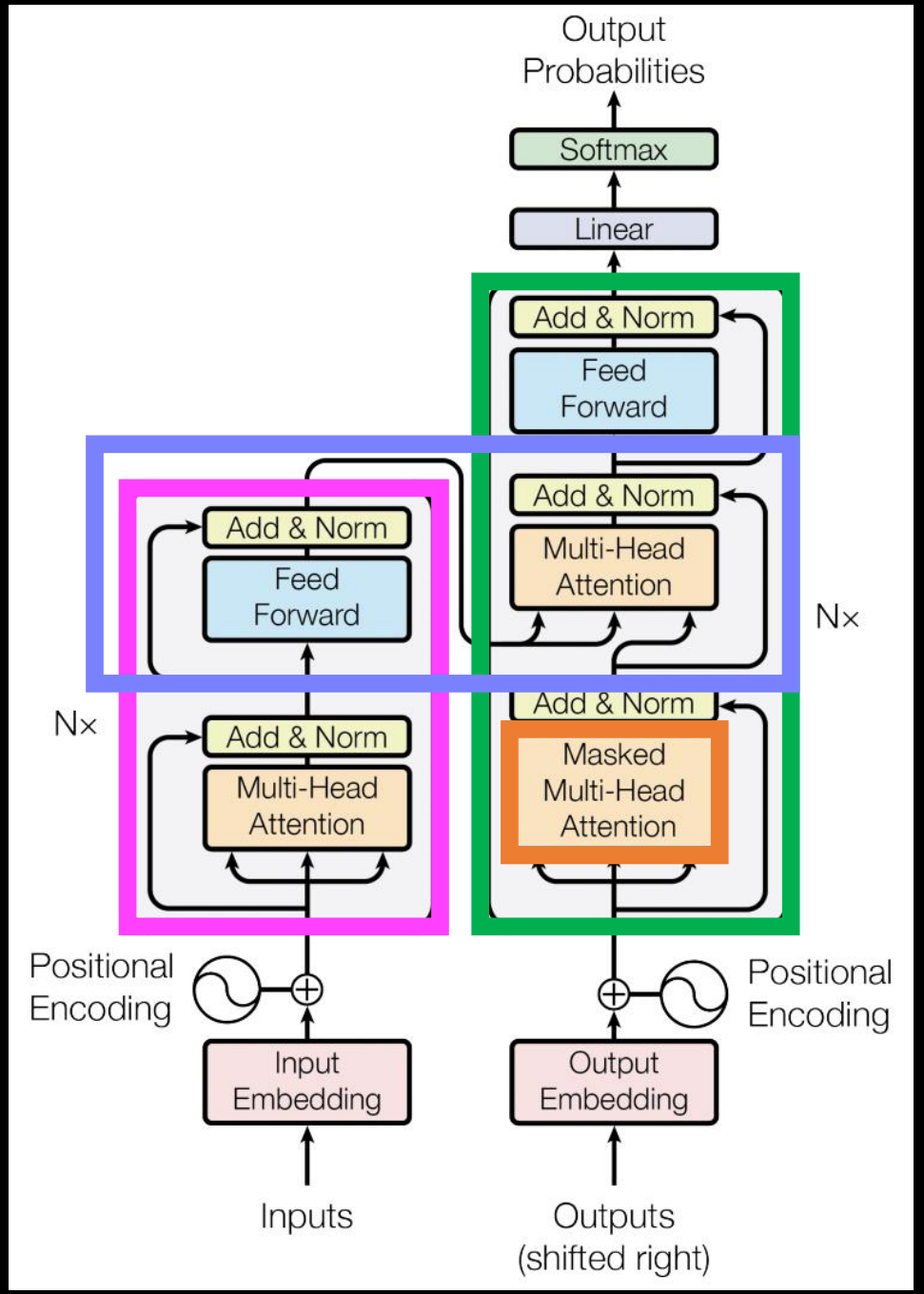
Overview: Autoregressive Model



http://jalammar.github.io/illustrated-transformer/

Architecture

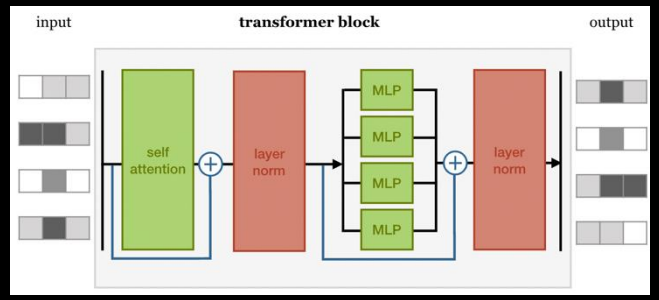
- Encoder (provides context for translation)
- Decoder (translates)
- Nx = 6 chained blocks (encoder & decoder)
- Cross-attention: enables decoder to attend to all inputs (i.e., keys and values come from the encoder and query comes from the decoder)
- Masking: cannot see future predictions by setting future values to $-\infty$, which softmax turns to 0



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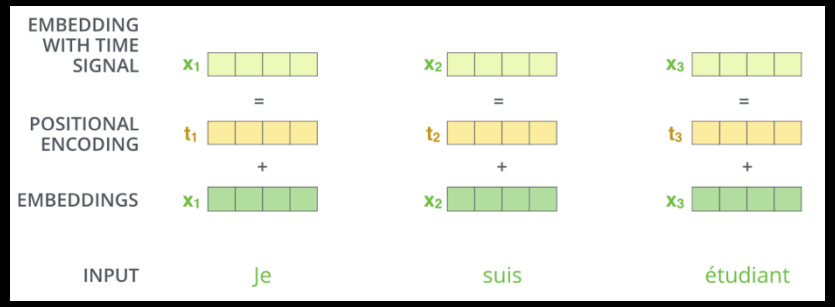
Architecture: Input/Output

- Input embedding matrix values for creating a dense vector are tied to output matrix for recovering one-hot word
- Positional encoding reveals order of sequence tokens
  - Addresses the lack of sensitivity to token order: shuffling order of input tokens yields same outputs in shuffled order

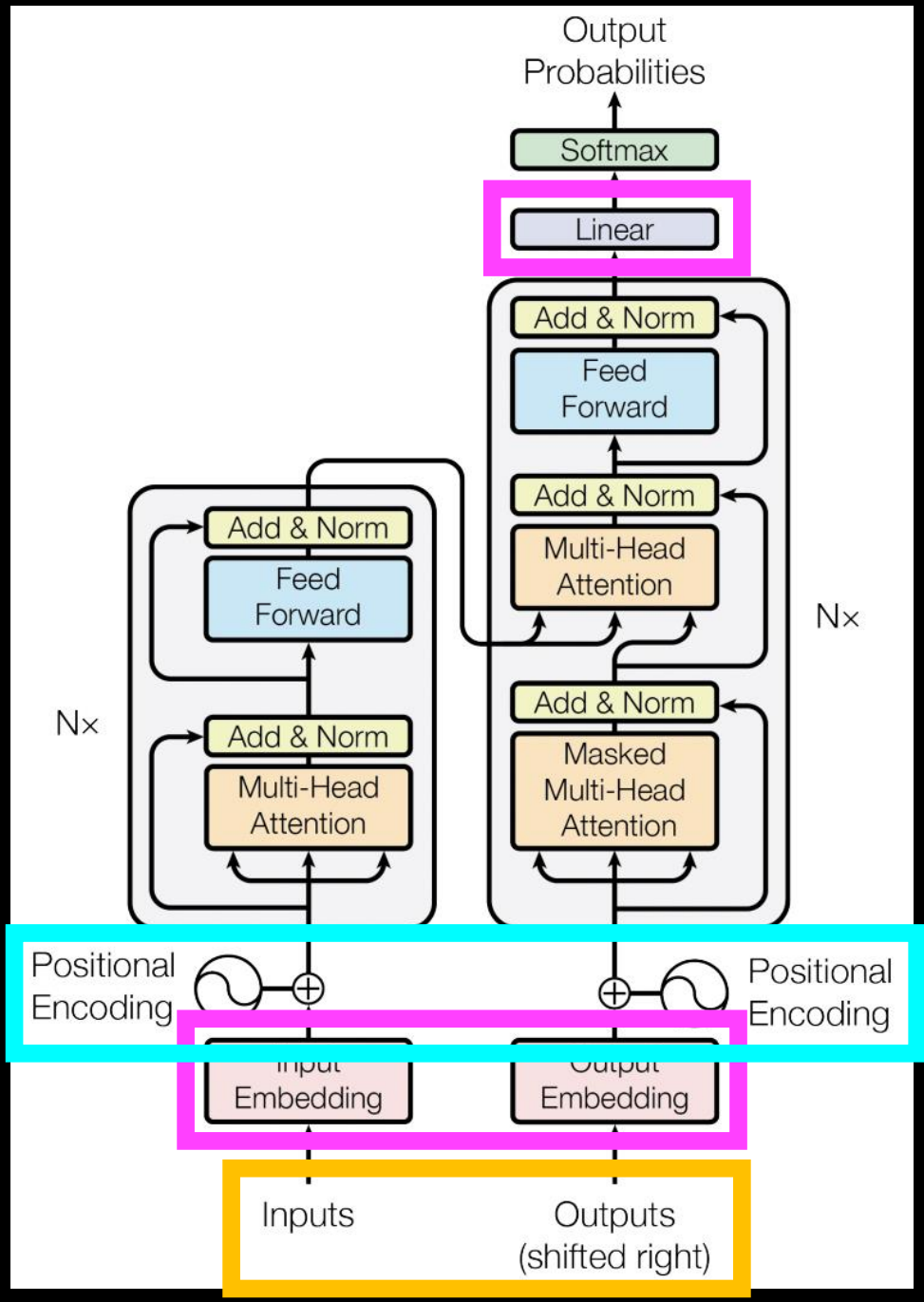


http://peterbloem.nl/blog/transformers

  - Approach: hard-coded and learning approaches successful

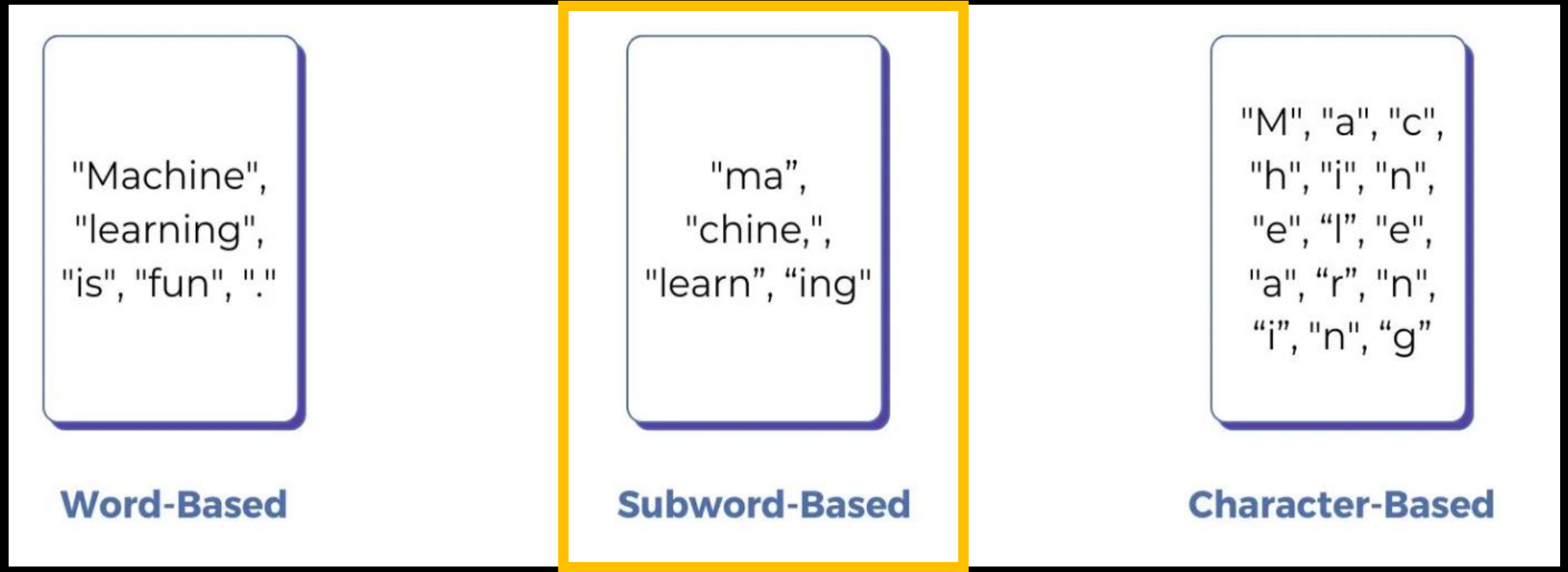


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Architecture: Input Encoding Options



https://eastgate-software.com/jp/what-is-tokenization-in-nlp-everything-you-need-to-understand/

What are its limitations?

- Huge vocabulary: ~1M words in English alone
- Novel words: information lost since one token ("UNK") for all instances

Middle ground, widely used in modern neural networks

What are its limitations?

Reduced semantics: greater meaning ambiguity for each character (e.g., "a")

Longer input: greater computational expense; e.g., 17 characters vs 4 words

Subword-Based: Byte-Pair Encoding

1. Identify all tokens in the training data with their frequency
2. Define vocabulary size; e.g., 14
3. Add all characters in the tokenized input to the vocabulary; e.g.,

Character sequence	Cost
C o s t	2
b e s t	2
m e n u	1
m e n	1
c a m e l	1

https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

Subword-Based: Byte-Pair Encoding

1. Identify all tokens in the training data with their frequency
2. Define vocabulary size; e.g., 14
3. Add all characters in the tokenized input to the vocabulary; e.g.,
4. Until vocabulary is filled, add merged highest frequency symbol pairs (cost = frequency)

e.g., What are the highest frequency symbol pairs?

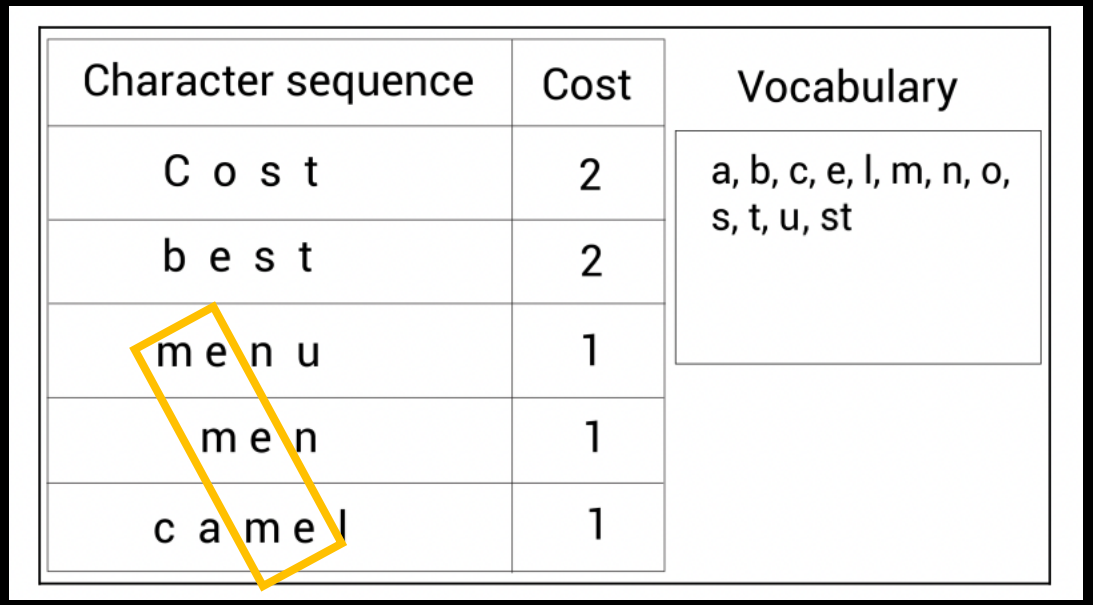
Character sequence	Cost	Vocabulary
C o s t	2	a, b, c, e, l, m, n, o, s, t, u
b e s t	2	
m e n u	1	
m e n	1	
c a m e l	1	

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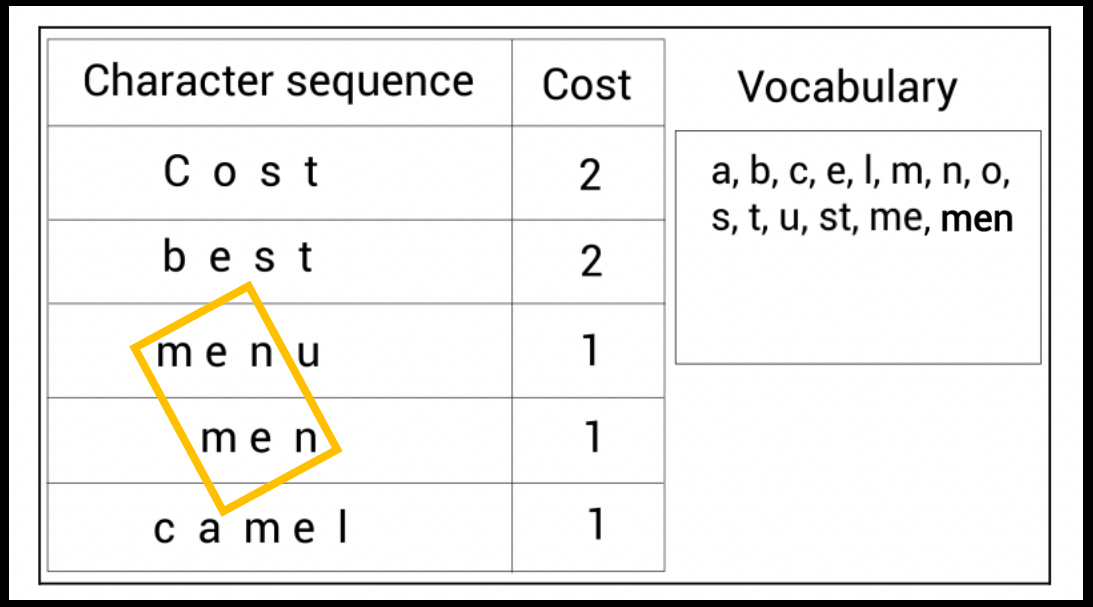
Vocabulary: a, b, c, e, l, m, n, o, s, t, u, st

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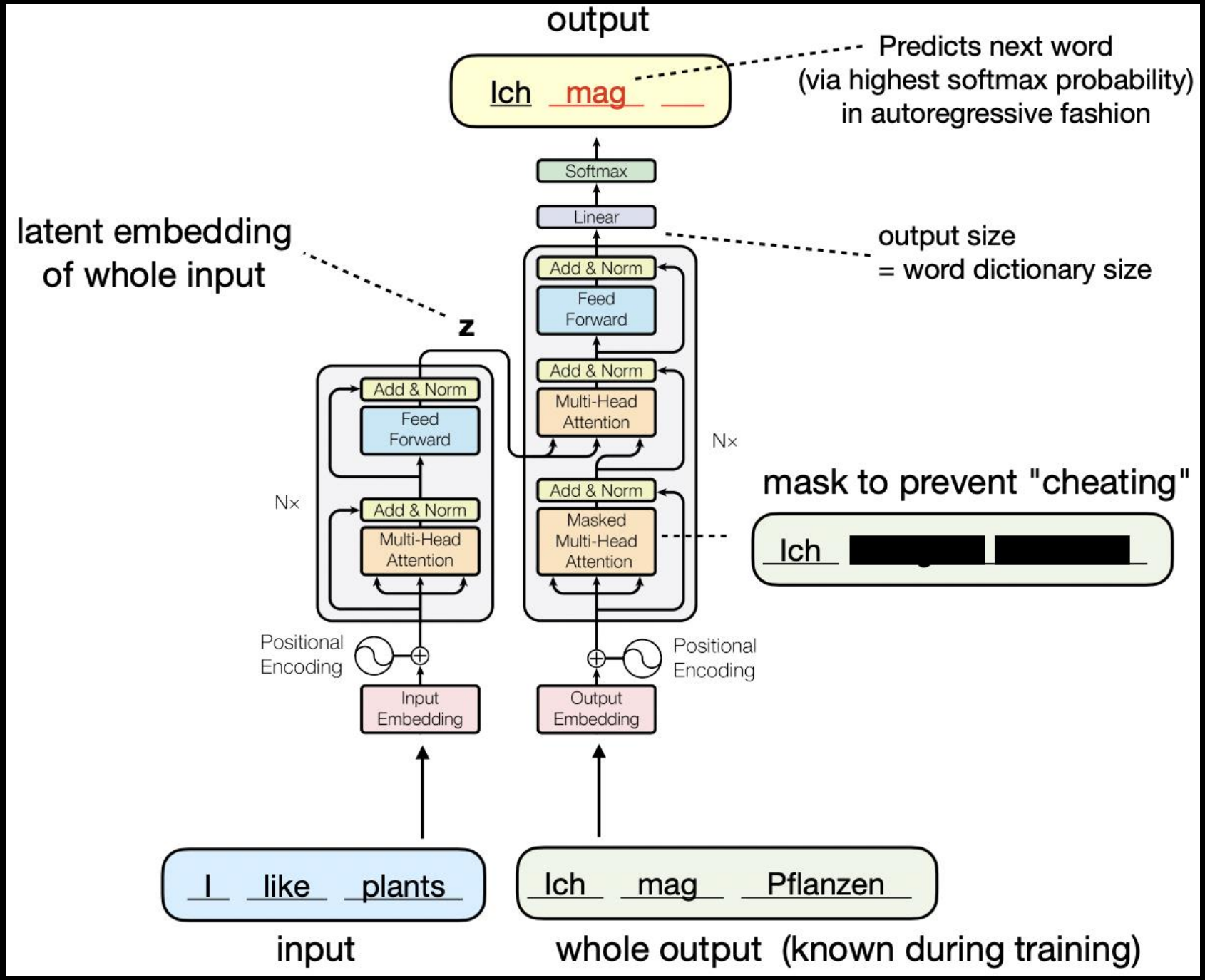


Character sequence	Cost
C o s t	2
b e s t	2
m e n u	1
m e n	1
c a m e l	1

Vocabulary
a, b, c, e, l, m, n, o, s, t, u, st, me, **men**

https://static.packt-cdn.com/downloads/9781838821593_ColorImages.pdf

Architecture: Example



https://sebastianraschka.com/pdf/lecture-notes/stat453ss21/L19_seq2seq_rnn-transformers__slides.pdf

Training: What's New

- Datasets: (1) English-to-German translation with ~4.5M sentence pairs and ~37,000 tokens; (2) English-to-French translation with ~36M sentence pairs and ~32,000 tokens
- Weight initialization: unspecified (implementation set norm of Glorot/Xavier method)

$$\left[-\frac{\sqrt{6}}{\sqrt{n_j + n_{j+1}}}, \frac{\sqrt{6}}{\sqrt{n_j + n_{j+1}}}\right]$$

fan in*:* # of neurons entering layer *j+1*

fan out*:* # of neurons leaving layer *j+1*

- Optimizer: Adam
- Batch size: 25,000 pairs of source and target tokens
- Regularization: dropout at a subset of layers
- Beam search: beam width of 4
- HW: 8 NVIDIA P100 GPUs (up to 3.5 days to train)

Experiments: Transformer Improves Performance While Also Training Faster!

Model	BLEU		Training Cost (FLOPs)	
	EN-DE	EN-FR	EN-DE	EN-FR
ByteNet [18]	23.75			
Deep-Att + PosUnk [39]		39.2		$1.0 \cdot 10^{20}$
GNMT + RL [38]	24.6	39.92	$2.3 \cdot 10^{19}$	$1.4 \cdot 10^{20}$
ConvS2S [9]	25.16	40.46	$9.6 \cdot 10^{18}$	$1.5 \cdot 10^{20}$
MoE [32]	26.03	40.56	$2.0 \cdot 10^{19}$	$1.2 \cdot 10^{20}$
Deep-Att + PosUnk Ensemble [39]		40.4		$8.0 \cdot 10^{20}$
GNMT + RL Ensemble [38]	26.30	41.16	$1.8 \cdot 10^{20}$	$1.1 \cdot 10^{21}$
ConvS2S Ensemble [9]	26.36	**41.29**	$7.7 \cdot 10^{19}$	$1.2 \cdot 10^{21}$
Transformer (base model)	27.3	38.1	**$3.3 \cdot 10^{18}$**	
Transformer (big)	**28.4**	**41.8**	$2.3 \cdot 10^{19}$	

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The End